Medical Physics

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Chapter 9: Optics

Chapter 9: Optics

You will learn:

1. The laws that govern the reflection and refraction of light.
2. The circumstances under which light is totally reflected at an interface.
3. Why concave and convex mirrors form different kinds of images
4. What aspects of a lens determine the type of image it produces.
5. What causes various defects in human vision, and how they can be corrected.
6. The principle of the simple magnifier.
7. How microscopes and telescopes work.
8. What happens when two waves combine, or interfere, in space.
9. How to understand the interference pattern formed by the Interference of two coherent light.
10. How to calculate the intensity at various points in an interference pattern.
11. How interference occurs when light reflects from the surface of a thin film.
12. How interference makes it possible to measure extremely small distances.
13. How to understand the diffraction pattern formed when coherent light passes through a narrow slit.
14. How to calculate the intensity at various points in a single-slit diffraction pattern.
15. How scientists use diffraction gratings for precise measurements of wavelength.
16. How x-ray diffraction reveals the arrangement of atoms *in* a crystal.
17. How diffraction sets limits on the smallest details that can be seen with a telescope.

Chapter 9: optics

Laws of Reflection

$$\theta = \theta'$$

Law of Refraction

Snell's Law

$$n1 sin\theta 1 = n2 sin\theta 2$$

For a thin lens

$$\frac{1}{p} + \frac{1}{q} = \frac{1}{f} \qquad M = \frac{h'}{h} = -\frac{q}{p}$$

Young's double-slit experiment

$$\delta = r_2 - r_1 = \mathrm{d}\sin\theta$$

Michelson interferometer

$$N = \frac{\Delta L}{\lambda/2} = \frac{2\Delta L}{\lambda}$$

Fresnel diffraction

$$\mathrm{a}\sin\theta = m\lambda$$

m= ±1, ± 2,……

a: the width of the slit

Chapter 9: optics

Rayleigh's criterion

$$\theta_{min} = 1.22\frac{\lambda}{D}$$

The diffraction grating

$$d\sin\theta = m\lambda$$

m= 0, ±1, ± 2,……

d: *slit separation*

Bragg′s law： XRD

$$2d\sin\theta = m\lambda$$ **m= 1,2,3……**

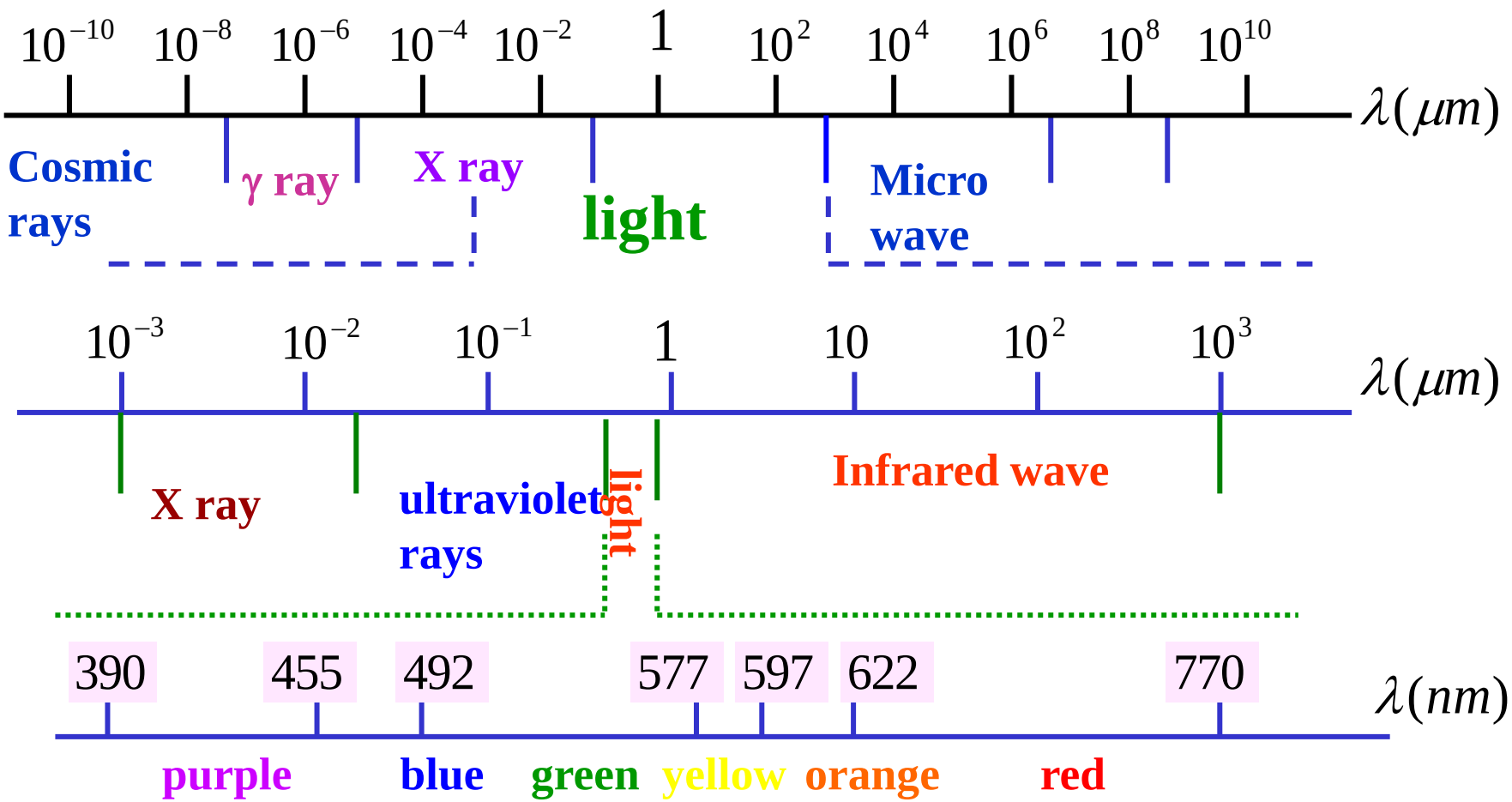
Malus' law

$$I = I_0 cos^2\theta$$

Brewster's Law

$$tan\theta_p = n_2/n_1$$

9.1 The Nature of Light： electromagnetic wave



9.1 The Nature of Light： electromagnetic wave

$$C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T\ m\ A^{-1}}$

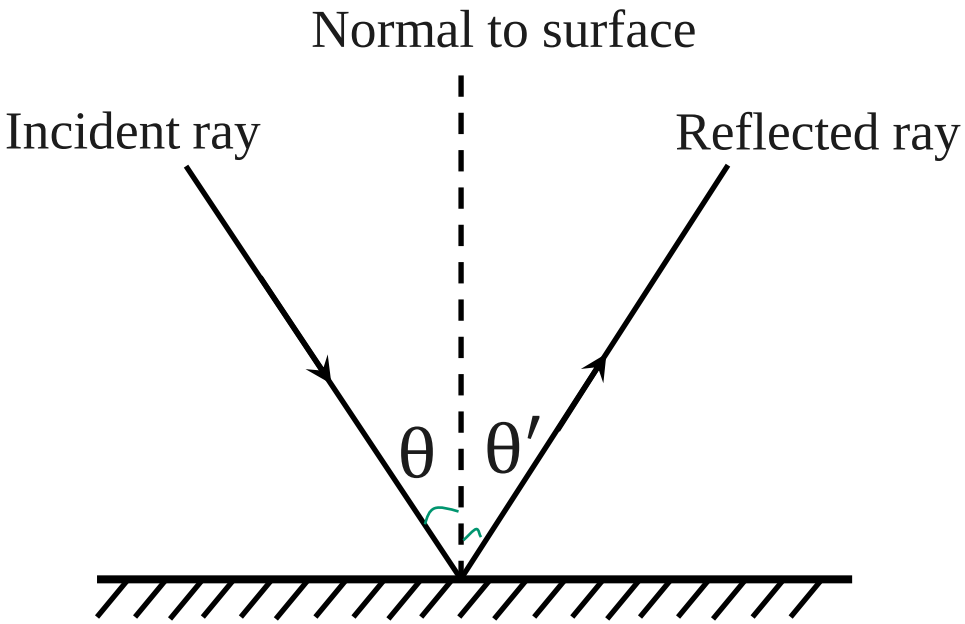
$\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2\ N^{-1}\ m^{-2}}$

$C = 2.998 \times 10^{8}\ \mathrm{m\ s^{-1}}$

The energy of a photon

$$E = h\nu$$

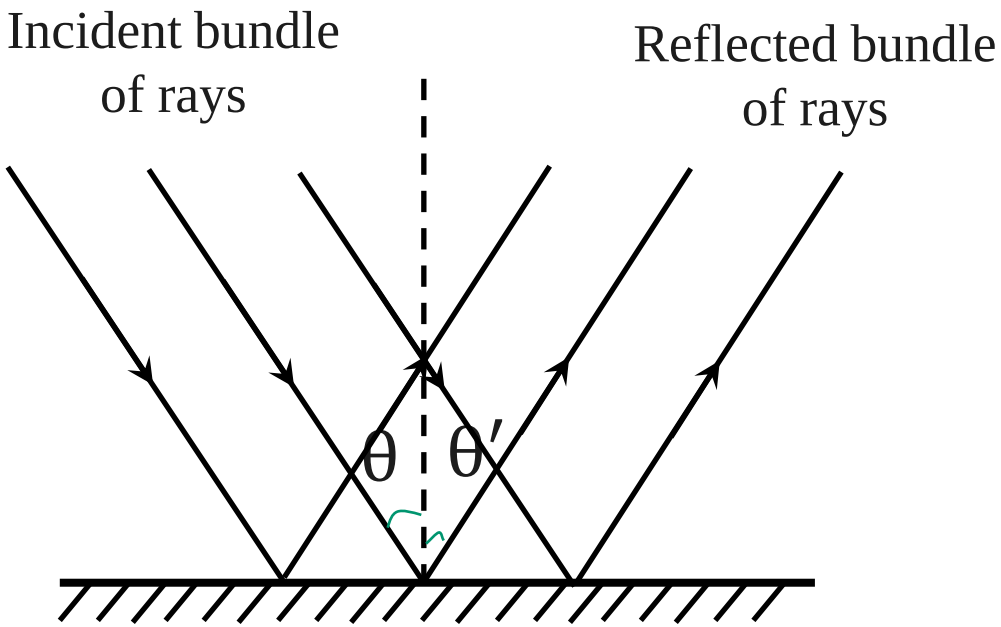
9-1: The Laws of Reflection and Refraction



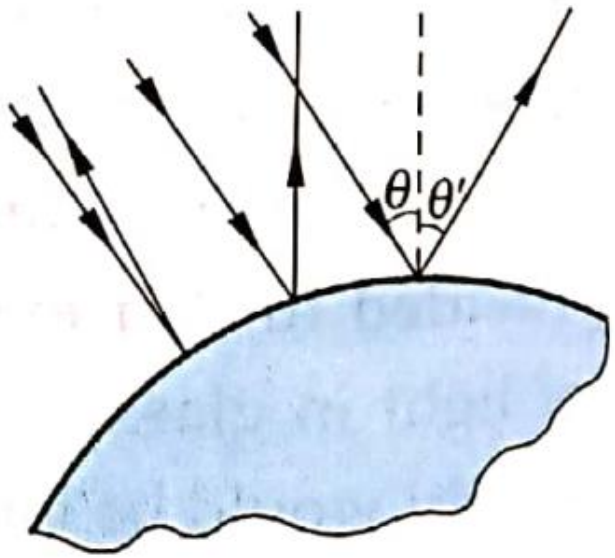
$$\theta = \theta'$$

9-1: Reflection of rays from various surfaces

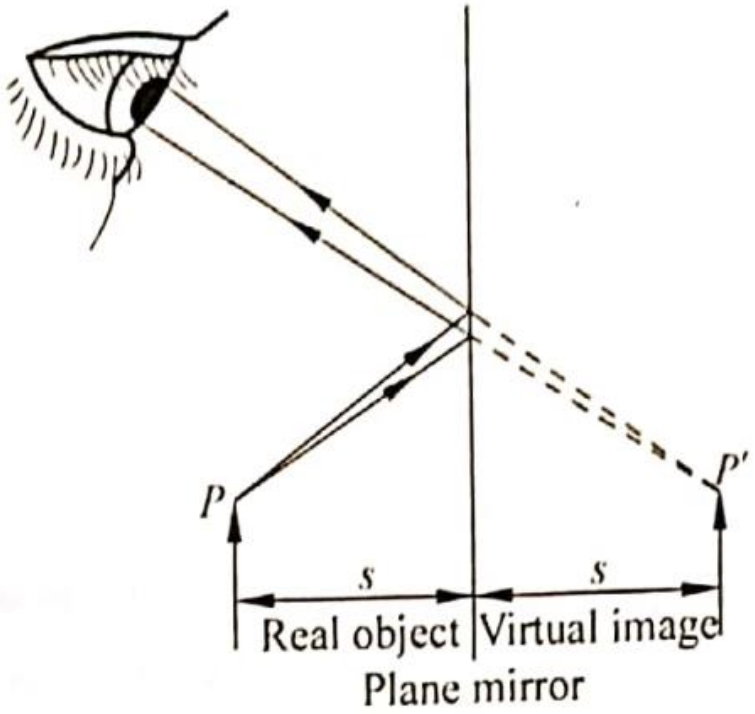
Flat surface



Curved surface

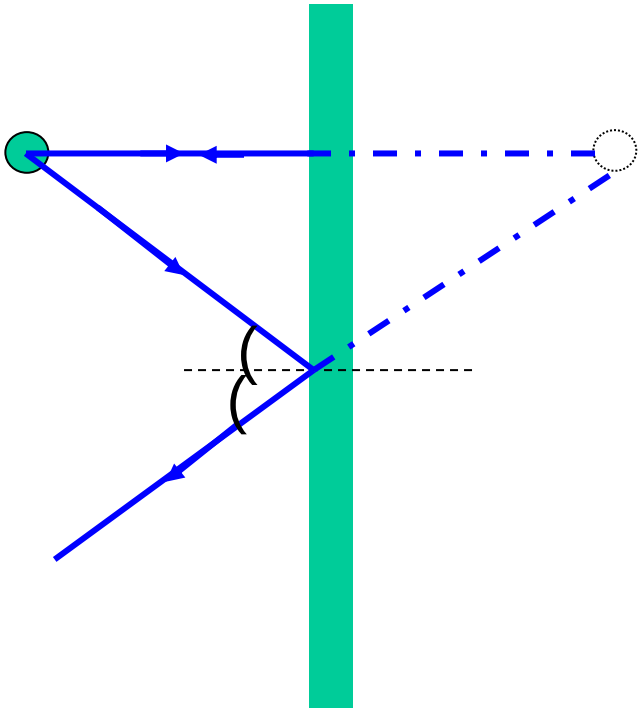


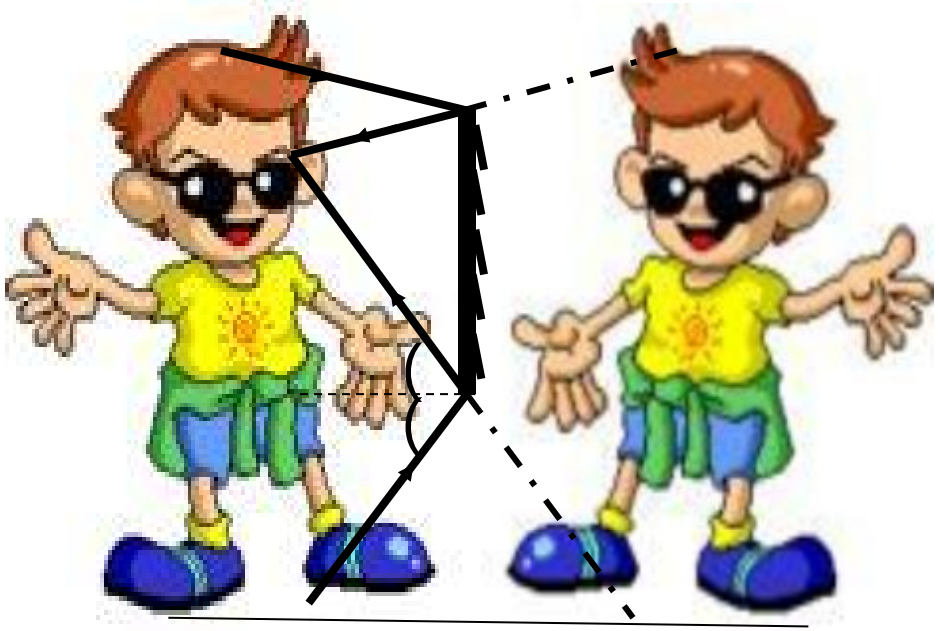
9-1: The Laws of Reflection and Refraction



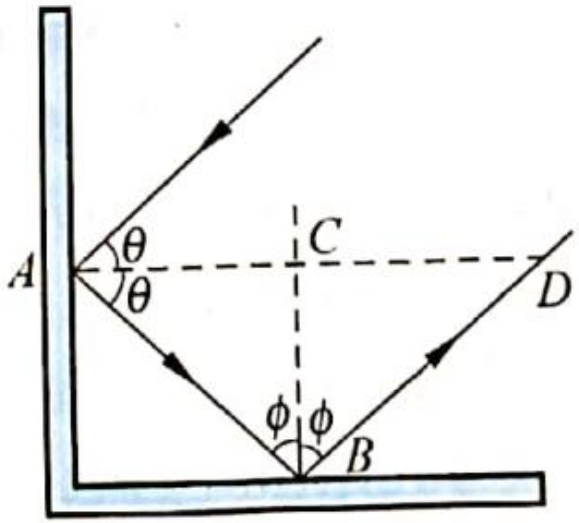
The upright arrow a distance s in front of the mirror is the object whose image appears behind the mirror. Two rays leave the point *P* at the tip of the object. These rays are reflected and enter the observer's eye. To the eye the rays appear to come from the point *F.* The point P' is the image of the point P. For a plane mirror the image appears to be as far behind the mirror as the object is in front of the mirror.

9-1： An example of the Reflection of Light: mirror



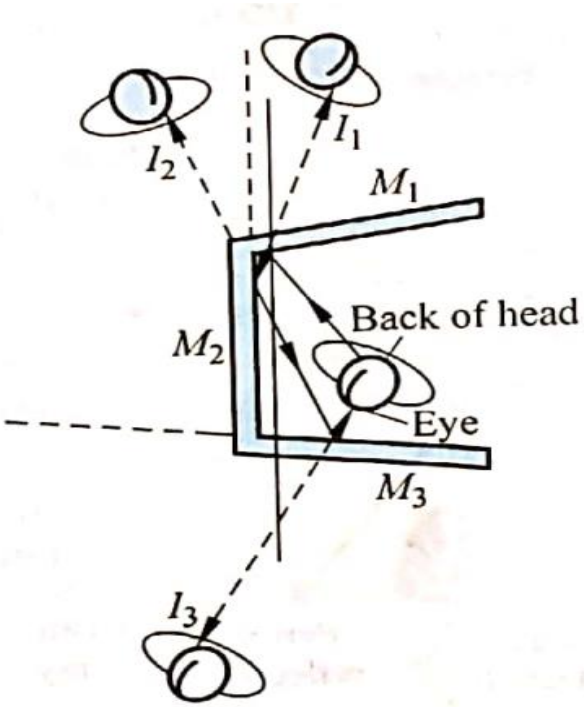


9-1: Example 1 of the reflection



A light ray is incident on two mirrors set at 90° with respect to each other. The incident ray is in the plane perpendicular to the two mirrors. Show that after two successive reflections, the outgoing ray will travel along the direction of incidence, no matter what angle the incident ray has with respect to the mirrors.

9-1: The Laws of Reflection and Refraction

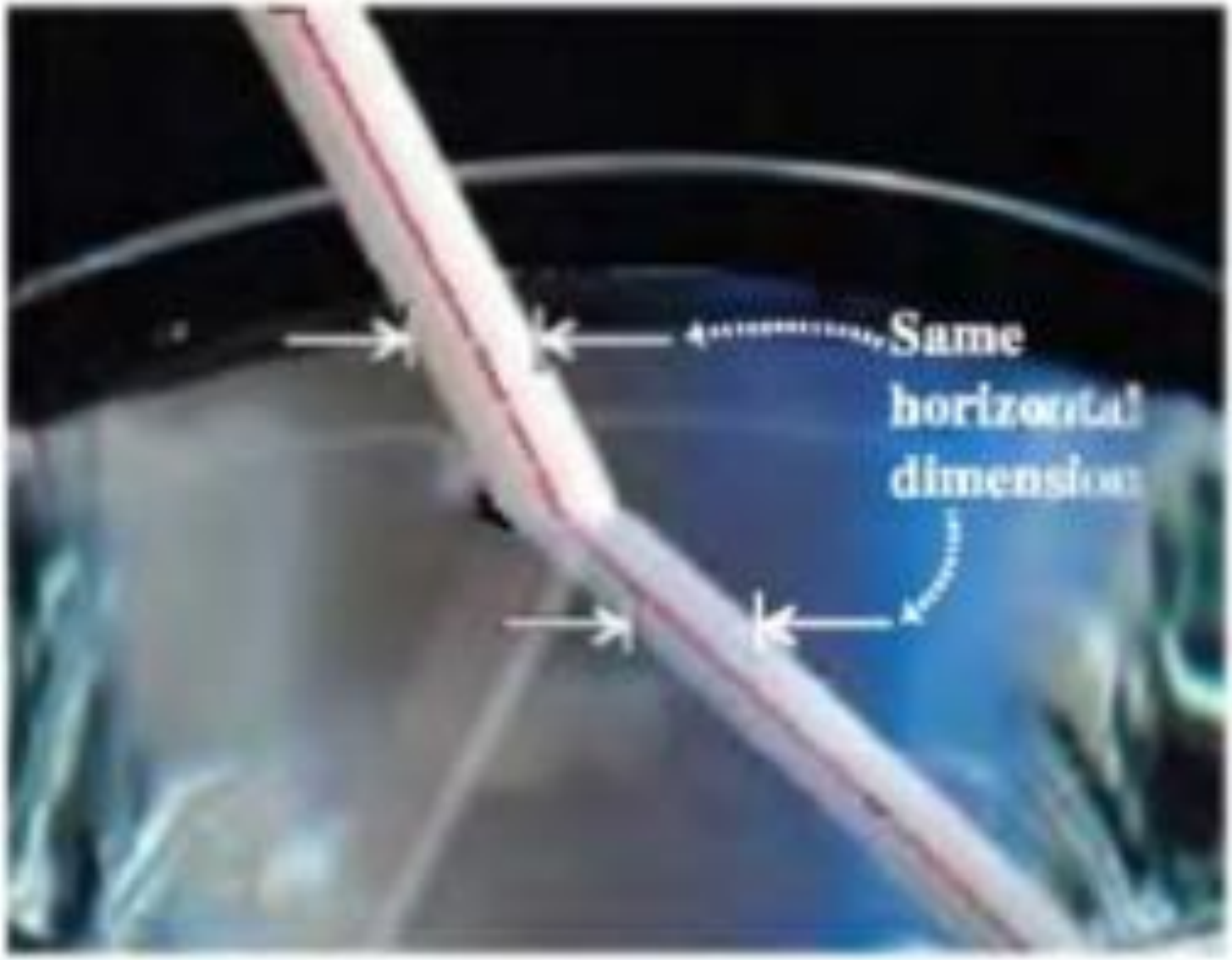


A triple mirror in a clothing store lets you see the back of your head by using three mirrors.

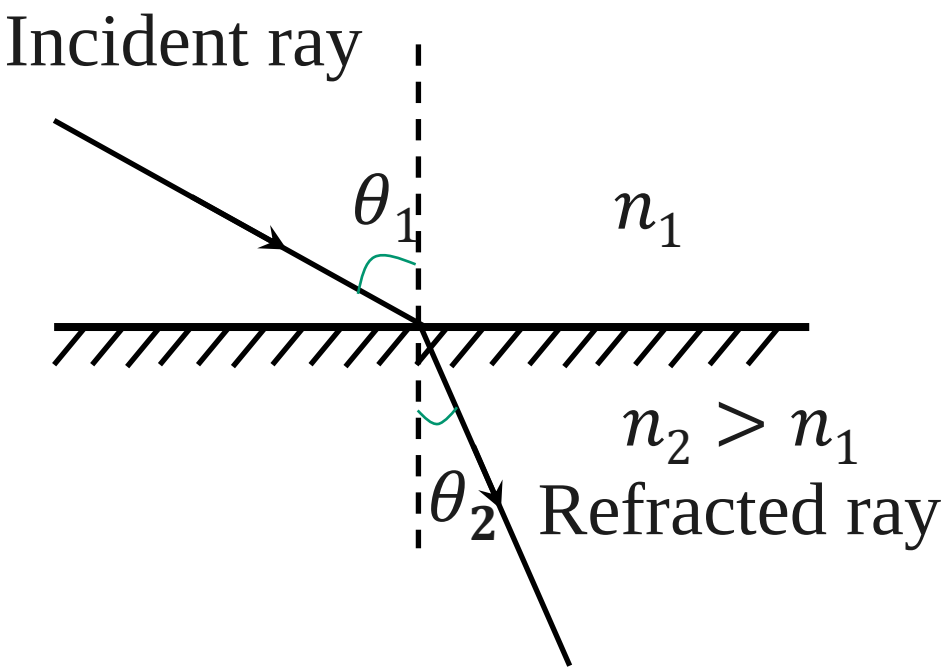


Multiple reflections from nearly parallel mirrors.

9-1：An example of the Refraction of Light



9-1: The Law of Refraction of Light

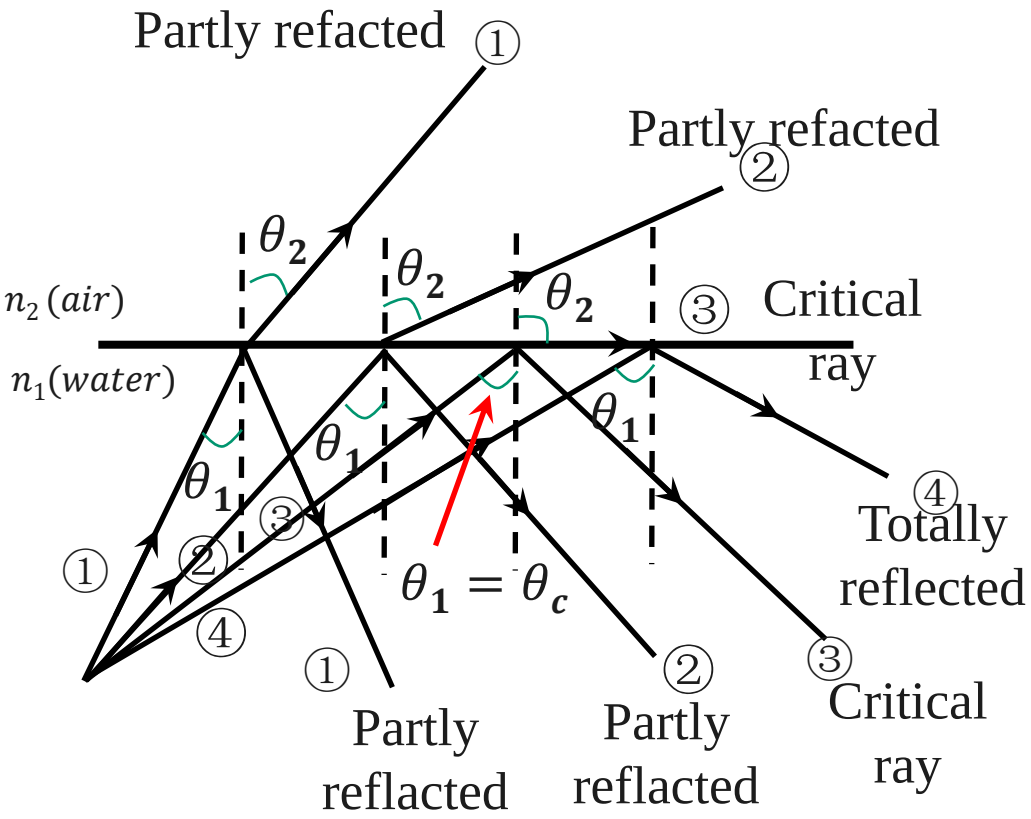


Snell's Law $n_1 sin\theta_1 = n_2 sin\theta_2$

9-1：Refraction index of some media

Media	Refraction index
Air	1.00029
water	1.333
Glass	1.468-1.755

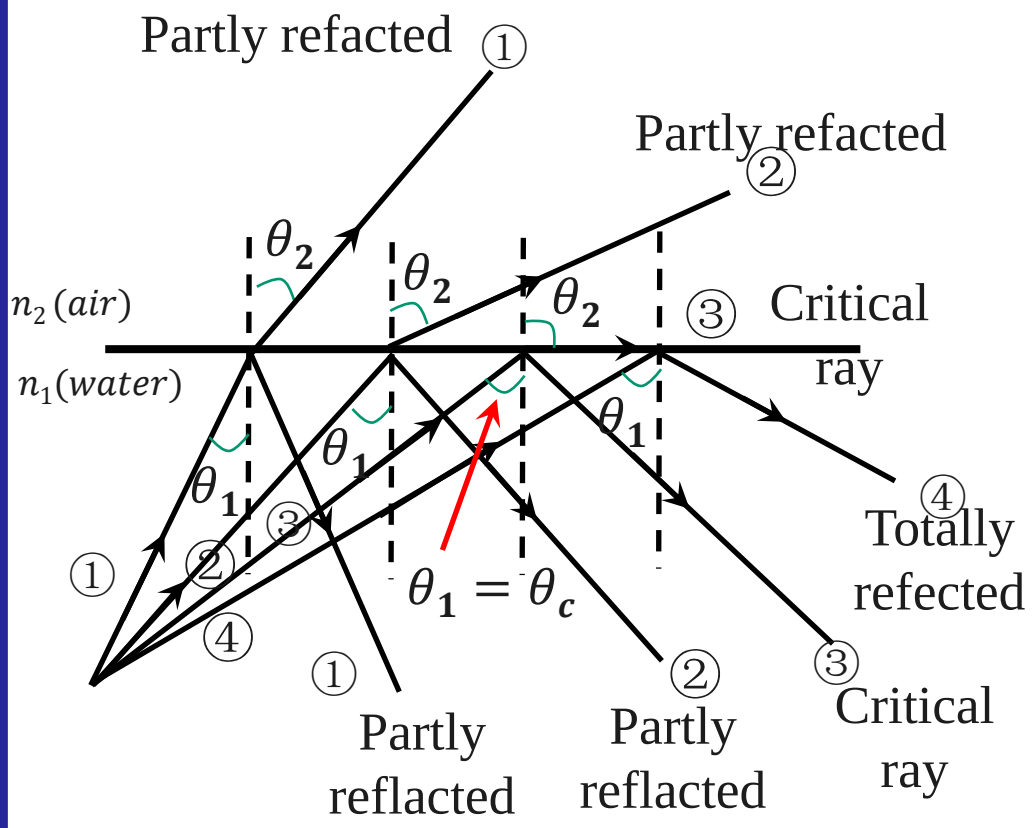
9-1： Total internal reflection



When the light from water to air, n2=1, $\theta 2 > \theta 1$

If $\theta_2 \geq 90^o$, what will happen?

9-1： Total internal reflection



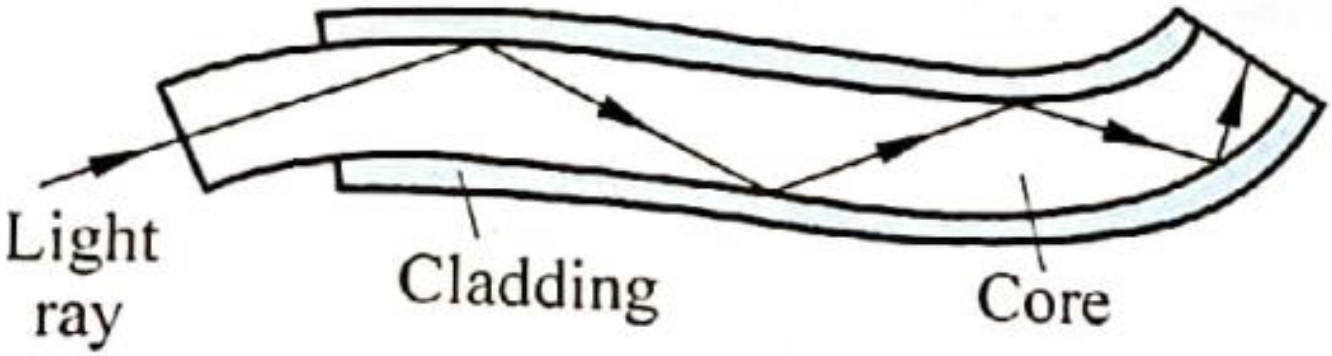
$$n_1 sin\theta_1 = n_2 sin\theta_2$$

$$sin\theta_1 = \frac{n_2}{n_1} sin\theta_2$$

$$sin\theta_1 = \frac{n_2}{n_1}$$

All rays are reflected: the reflection is total

9-1：Application of the total reflection

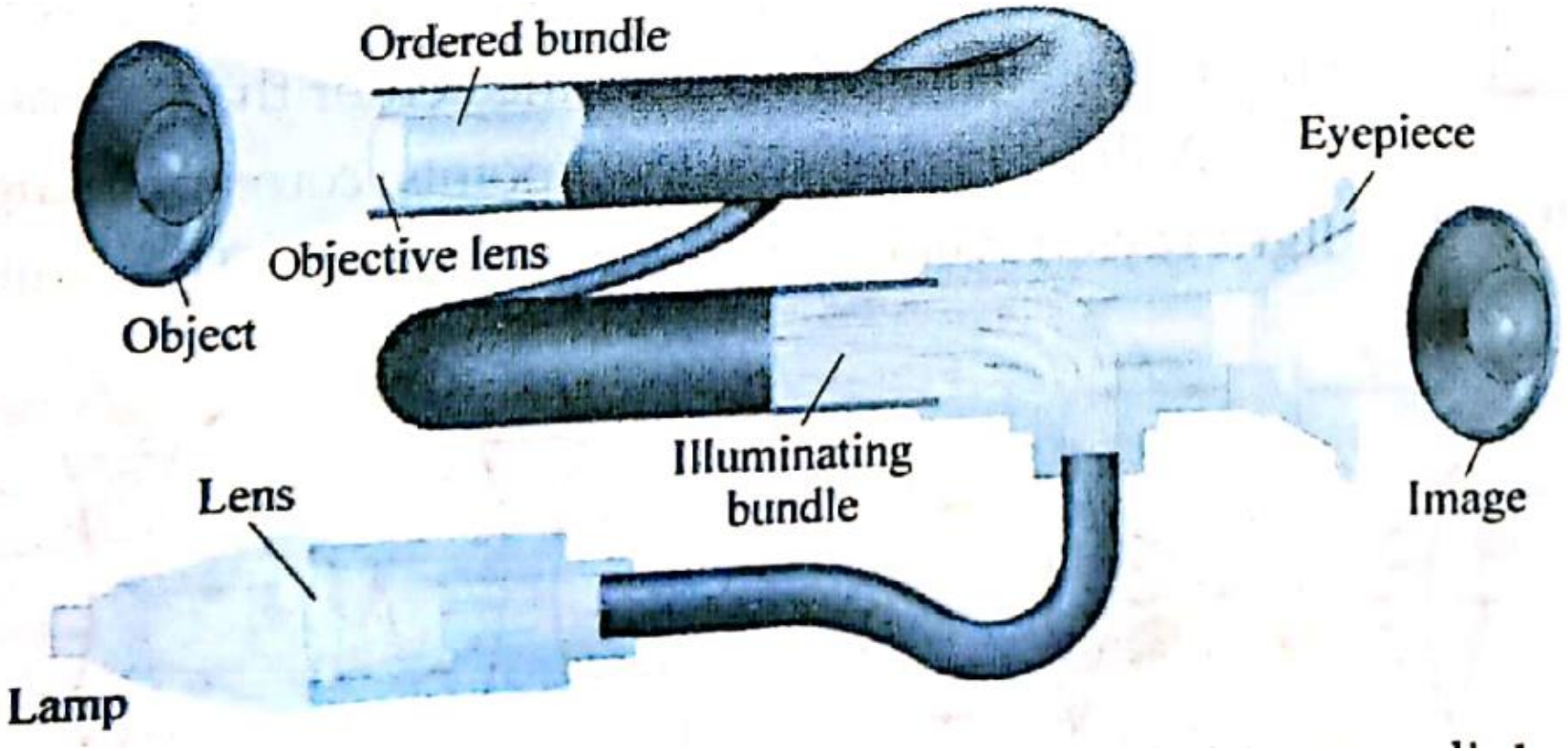


An important application of total internal reflection occurs in fiber optics, where hair-thin threads of glass or plastic, called optical fibers, “pipe” light from one place to another. Here is an optical fiber consists of a cylindrical. Inner core that carries the light and an outer concentric shell, with lower index of refraction, called cladding. The core is made from transparent glass or plastic that has a relative high index of refraction. Light enters one end of the core, strikes the core-cladding interface at an angle of incidence greater than the critical angle, and the light is reflected back into the core. Therefore, the light is travels inside the fiber along a zigzag path.

9-1： Optical fiber

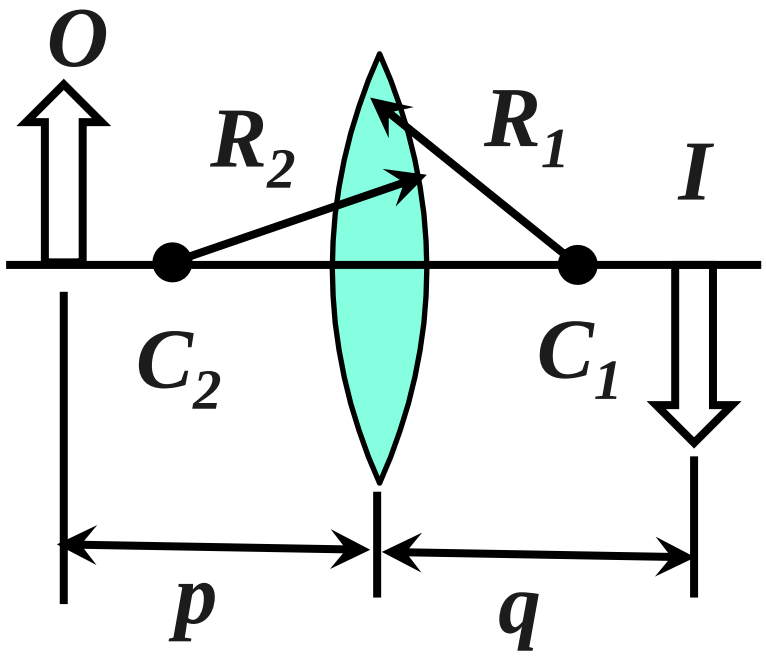


9-1：Application of the total reflection



A fiberscope contains two bundles of optical fibers: one set to carry light from a source to the area to be viewed, and another to carry light from the viewing area to the eye. The the doctor can see a polyp in the stomach

9.2 Thin lenses

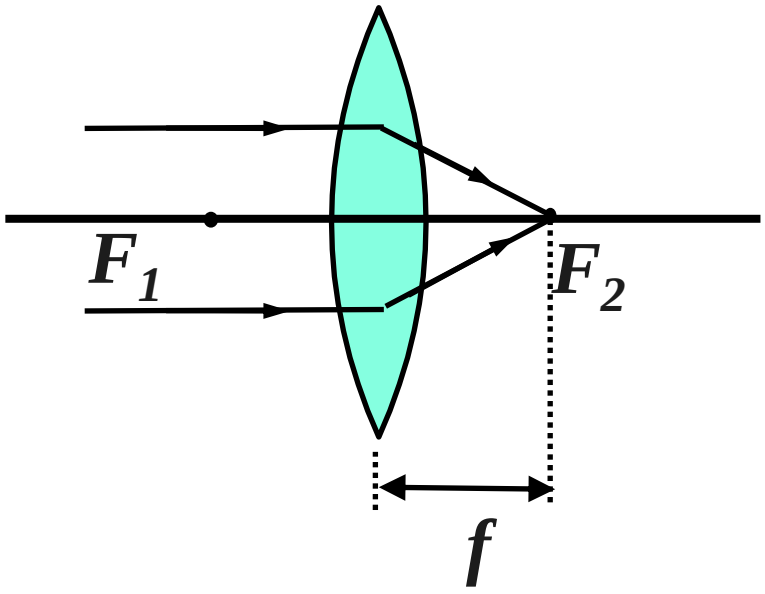


$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$$

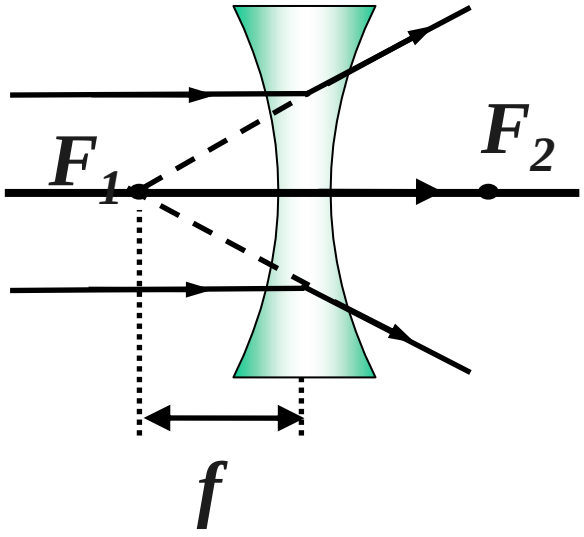
$$\frac{1}{f}=(n-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$

P: the distance of object
Q: the distance of image.
f: the focal length of lens.
O: object,
I: image

9.2 The object and image focal points



Biconvex lens
Converging lens



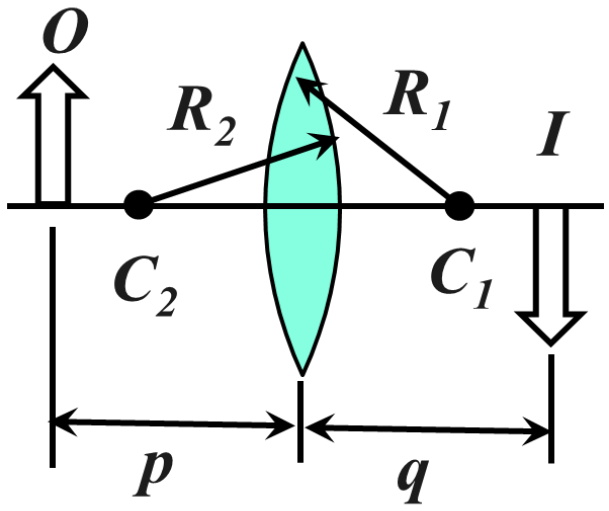
Biconcave lens
Diverging lens

9.2 How to determine the sign of p and q

Sign convention for thin lenses

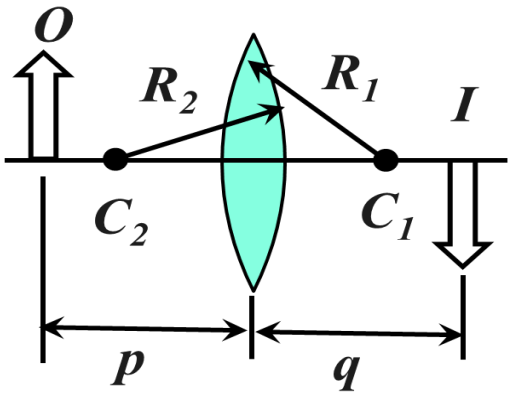
Front	Back
p: +	p: -
q: -	q: +
Incident light	Refraction light

p is + if the object is in front of the lens
p is - if the object is in back of the lens
q is + if the image is in back of the lens
q is - if the image is in front of the lens
R1, and R2 are + if the center of curvature is in back of the lens
R1 and R2 are - if the center of curvature is in front of the lens

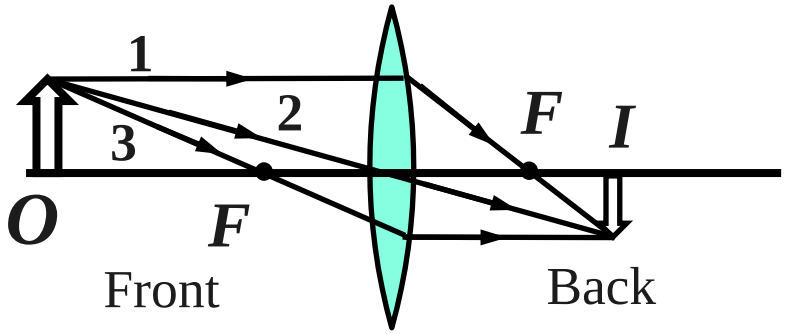


$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f}$$

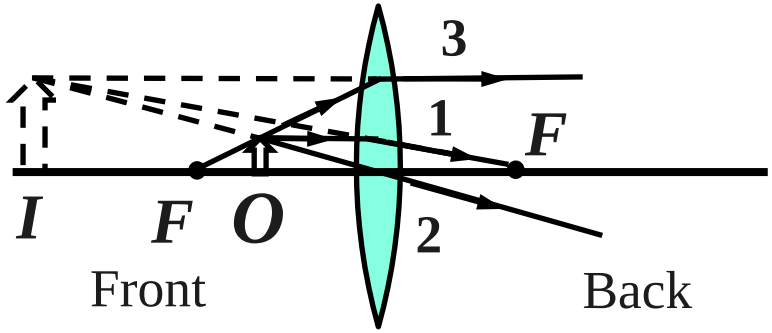
9.2 an example: a converging lens



$$\Downarrow \quad \frac{1}{p}+\frac{1}{q}=\frac{1}{f}$$



When p>f, p:+, q: +, R1: +, R2: -



When p<f, p:+, q: -, R1: +, R2: -

9.2 Thin lens

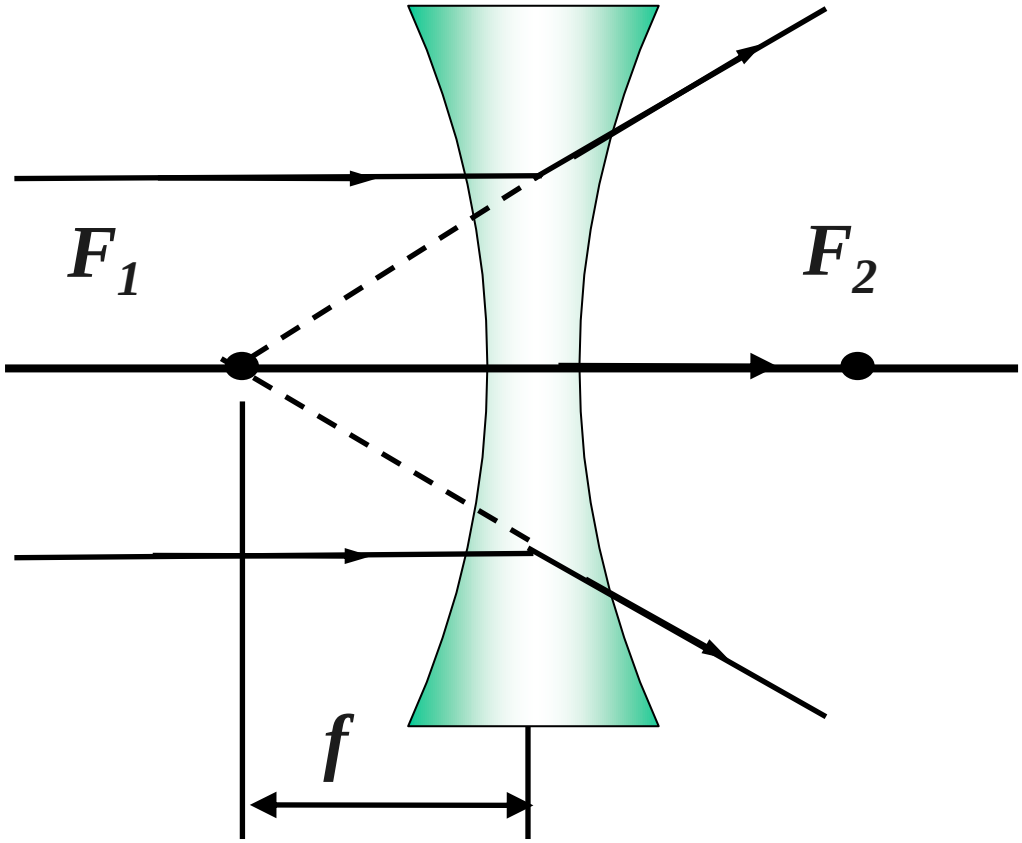
Consider a single thin lens illuminated by a real object (that the object is real means p>0). The lateral magnification of a thin lens is defined as the ratio of the image height h' to the object height h:

$$M = \frac{h'}{h} = -\frac{q}{p}$$

M: positive, the image is upright (erect) and on the same side of the lens as the object.
M: negative, the image is inverted and on the side of the lens opposite the object.

9.2 example

A diverging lens has a focal length of -20.0cm. An object 2.0cm. in height is placed 30.0cm in front of the lens. Locate the position of the image and find the magnification and the height of the image.



9.2 Example

Solution:

Using the thin lens Eq. (9-2) with p = 30.0cm and f=-20.0cm, we get

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f} \qquad \frac{1}{30}+\frac{1}{q}=\frac{1}{-20}$$

$$q = -12.0\ cm$$

The negative sign means the image is virtual. The magnification is

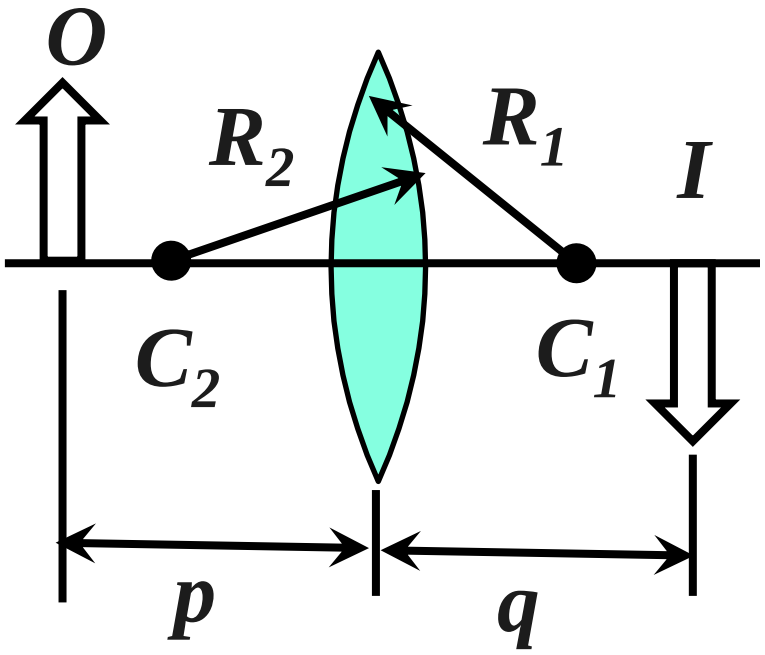
$$\mathrm{M}=\frac{\mathrm{h}'}{\mathrm{h}}=-\frac{\mathrm{q}}{\mathrm{p}}=-\frac{(-12.0\ \mathrm{cm})}{30.0\ \mathrm{cm}}=0.4$$

The height of the image is

$$\mathrm{h}'= Mh = (0.4)(2.0cm) = 0.8\ cm$$

9.2 Example 2

A converging lens of focal length 10.0cm forms an image of an object placed (a) 30.0cm, (b) 10.0cm, and (c) 5.0cm from the lens. Find the image distance and describe the image



9.2 Example 2

Solution: (a)

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f} \qquad \frac{1}{q}=\frac{1}{f}-\frac{1}{p}$$

$$\frac{1}{q}=\frac{1}{10}-\frac{1}{30} \qquad q=15\ cm$$

$$M=\frac{h\prime}{h}=-\frac{q}{p}=-\frac{(15\ cm)}{30.0\ cm}=-0.5$$

Thus, the image is reduced in size by one half, and the negative sign for M tells us that the image is inverted.

9.2 Example 2

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f} \qquad \frac{1}{q}=\frac{1}{f}-\frac{1}{p}$$

$$\frac{1}{q}=\frac{1}{10}-\frac{1}{10} \qquad q=\infty\,(infinity)$$

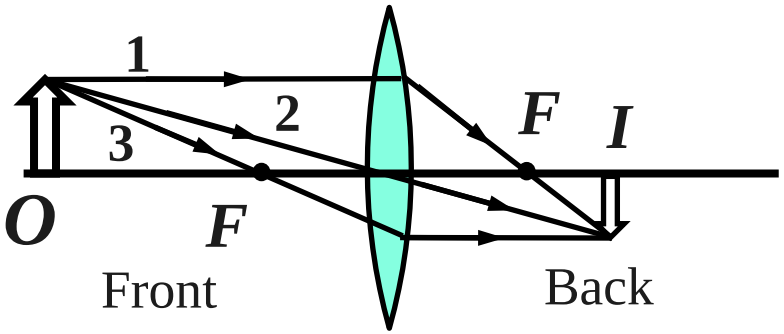
(C) p =5 cm

$$\frac{1}{q}=\frac{1}{10}-\frac{1}{5} \qquad q=-10\ cm$$

$$M=\frac{h\prime}{h}=-\frac{q}{p}=-\frac{-10\ cm}{5.0\ cm}=2.0$$

The negative image distance: the image is virtual.
The image is enlarged, and the positive sign for M tells us that the image is upright .

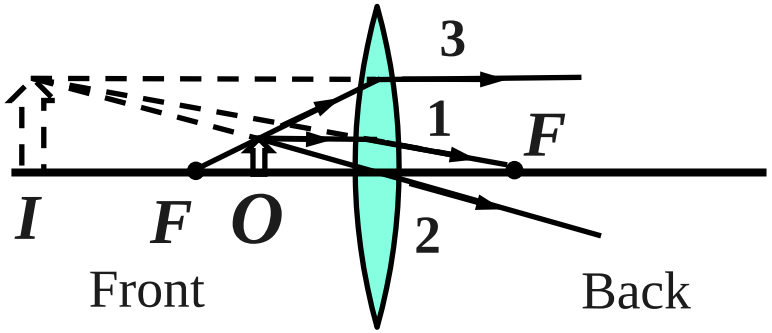
9.2 Example 2: for converging lens



$p = 30\ cm$
$q = 15\ cm$

$M =$**-0.5**

Image is real, reverted

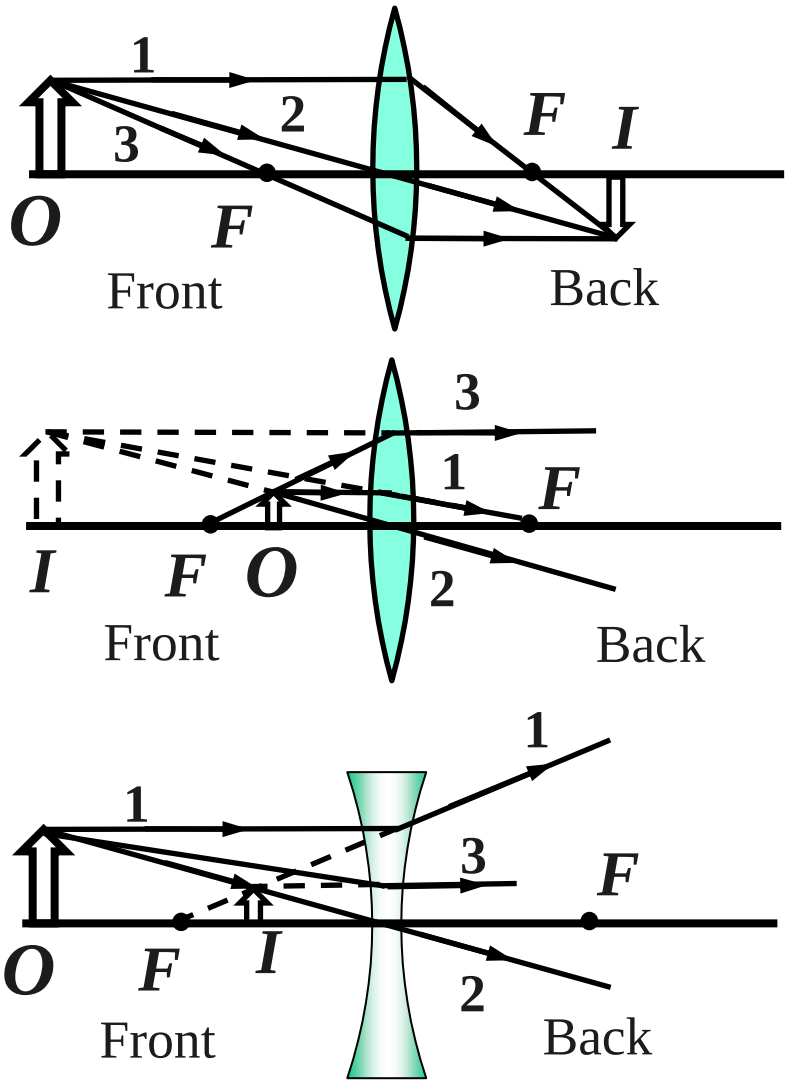


$p = 5\ cm$
$q = -10\ cm$

$M = 2$

Image is virtual, upright

9.2.2 Graphical Methods



Ray 1 is drawn parallel to the optical axis. After being refracted by the lens, this ray passes through (or appears to come from) one of the focal points.
Ray 2 is drawn through the center of the lens. This ray continues in a straightline.
Ray 3 is drawn through the object focal point and emerges from the lens parallel to the optical axis.

Homework

Page 218-220

Q: 2, 4, 7, 8

Homework file name:

Roll number+week 9 + name,

for example: 2023800901XX + week 9 + Alfa Robot

pdf file, send to the cloud)

9.2.3 The Eye

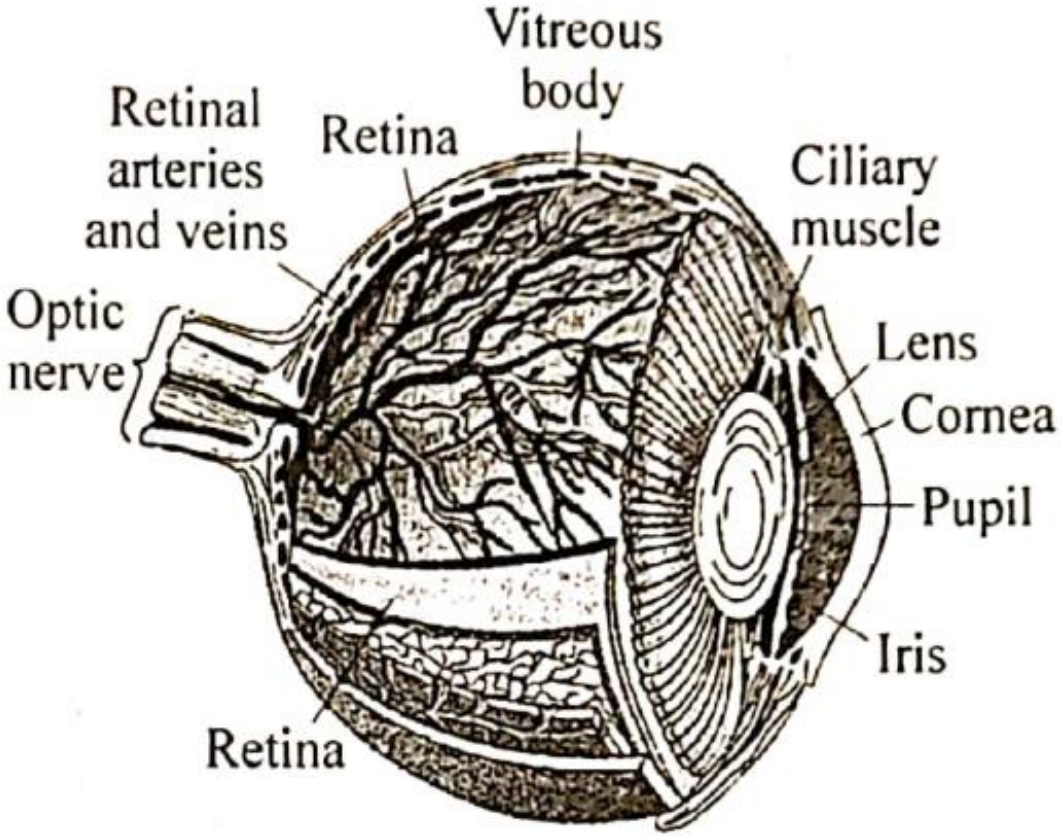
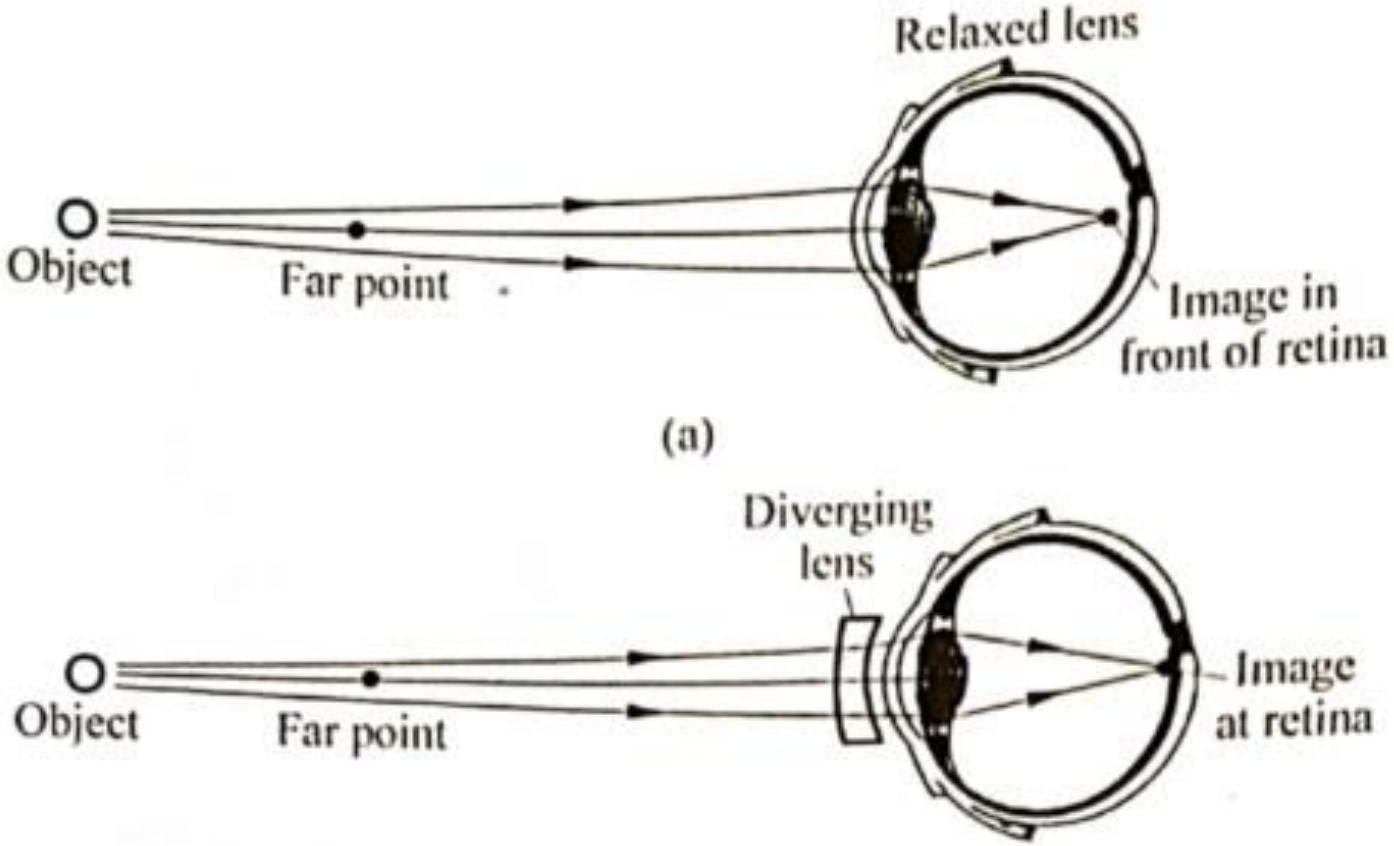


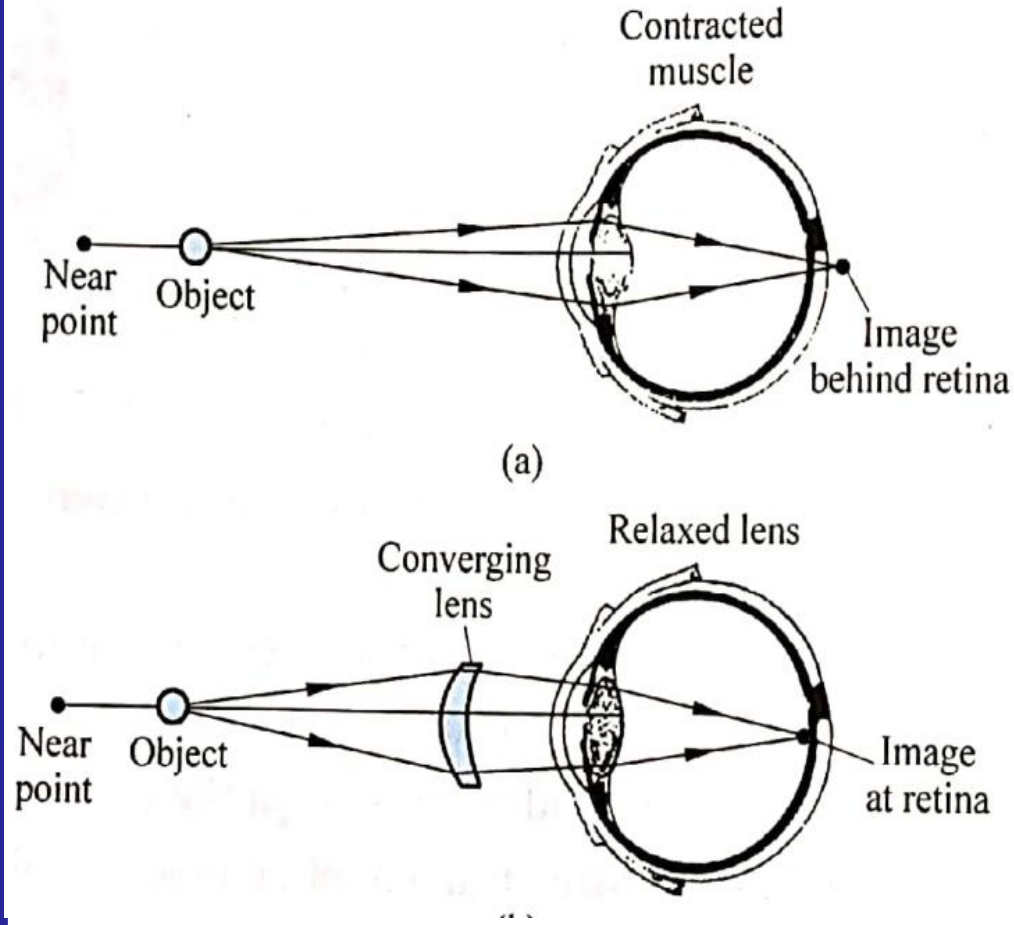
Fig. 9-15 Essential parts of the eye.

9.2.3 The Eye: nearsighted



(a)When a nearsighted person looks at a distant object that lies beyond the far point, the image is formed in front of the retina, resulting in blurred vision; (b) The condition of nearsightedness can be corrected with a diverging lens.

9.2.3 Farsighted



When a farsighted person looks at an object inside the near point, the image is formed behind the retina, resulting in blurred vision. The eye muscle contracts to try to bring the object into focus; (b) The condition of farsightedness can be corrected with a converging lens.

9.2.3 How to avoid nearsighted

Near point: the closest distance for which the lens of the relaxed eye can focus light on the retina.

10 years old: 18 cm,

20 years old: 25 cm

40 years old: 50 cm

60 years old: 500 cm

How to avoid the nearsight:

1. When you read or write, the distance from your eye to book is > 25 cm.
2. When you feel eye tired, look at far distance or do eye exercise to relax your eyes.
3. When you feel eye tired, do sports to relax your eyes.

9.2.3 An example

Example 1. (a) The near point of a certain eye is 100cm in front of the eye. What lens should be used to see clearly an object 25cm in front of the eye? (b) The far point of a certain eye is l m in front of the eye. What lens should be used to see clearly an object at infinity?

9.2.3 An example

Solution:

(a)Assume the image to be formed at the near point,

p = + 25cm, q= - 100cm

$$\frac{1}{p}+\frac{1}{q}=\frac{1}{f} \qquad \frac{1}{f}=\frac{1}{25\ cm}+\frac{1}{-100\ cm}$$

$f = 33.3\ cm$: *a converging lens* required

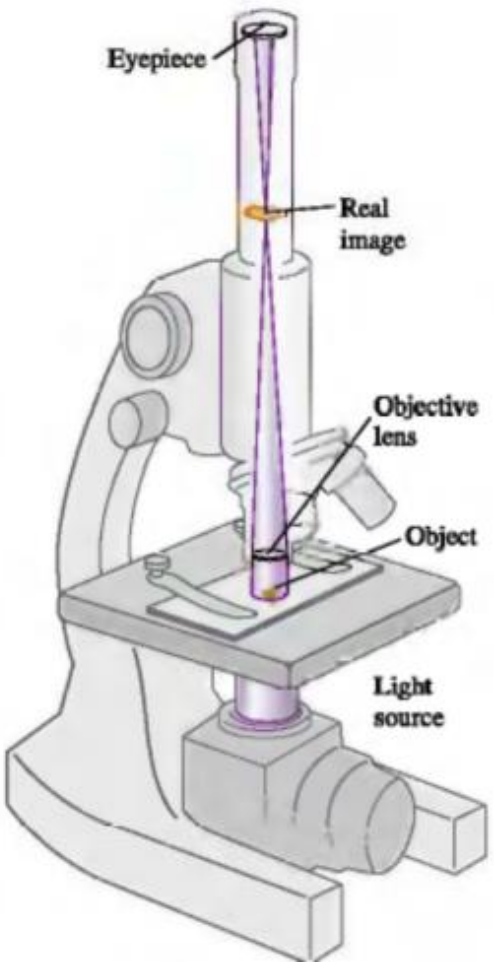
(b) Assume the image to be formed at the far point,

p = ∞, q= − 100cm $\qquad \frac{1}{f}=\frac{1}{\infty}+\frac{1}{-100\ cm}$

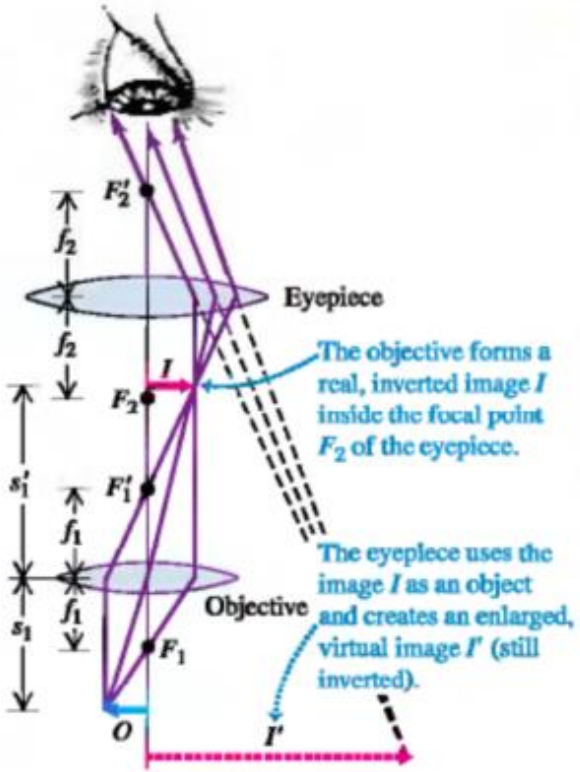
$f = -100\ cm$: *a diverging lens* required

9.2.4 Microscope and Telescope

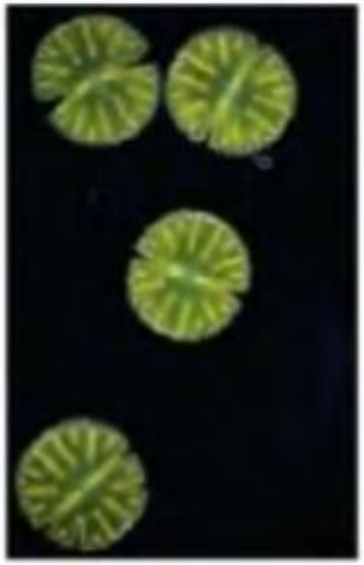
(a) Elements of a microscope



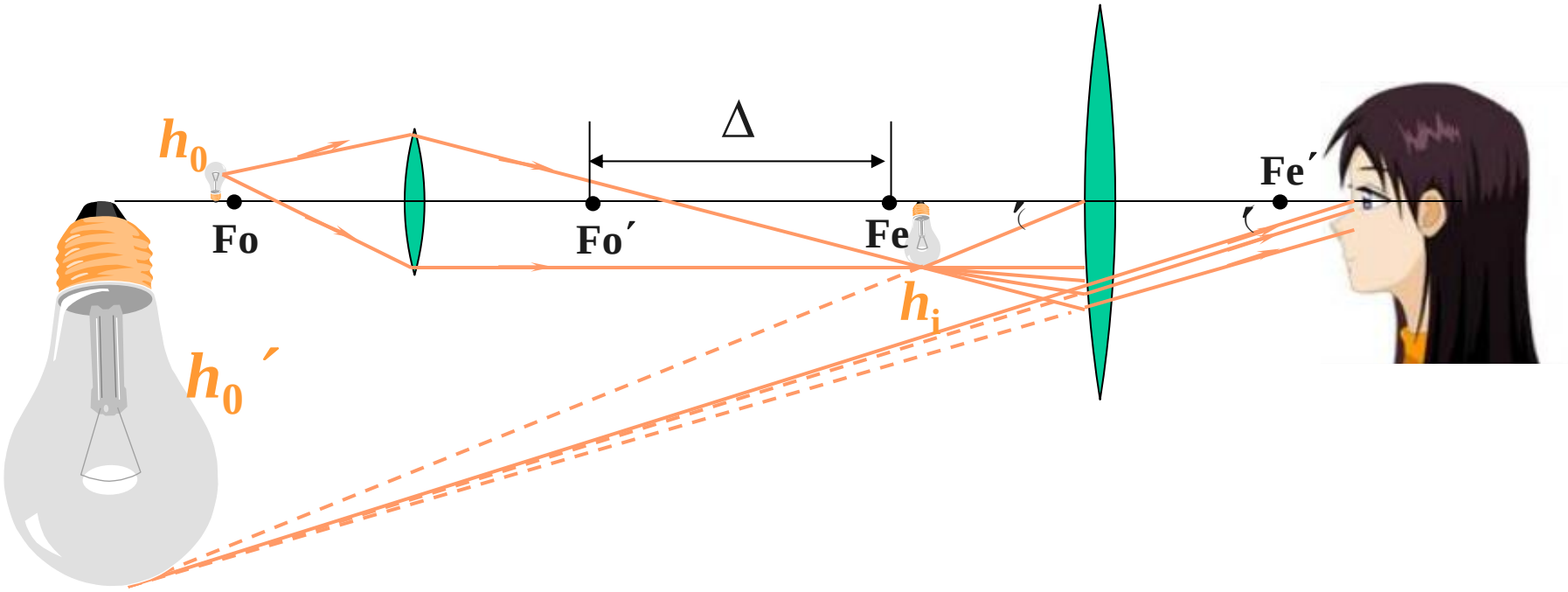
(b) Microscope optics



(c) Single-celled freshwater algae (*Micrasterias denticulata*)

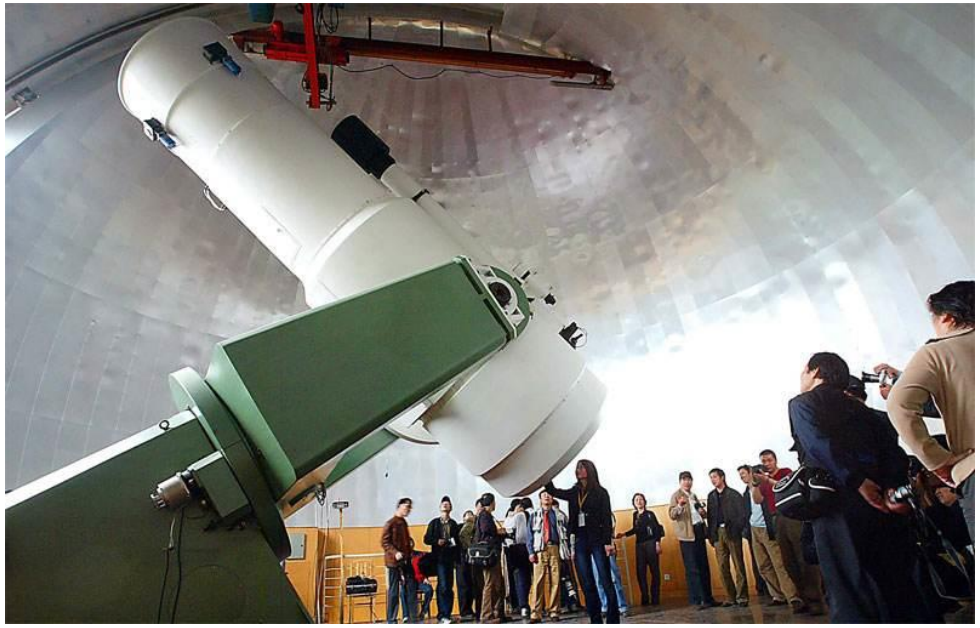


9.2.4 Microscope

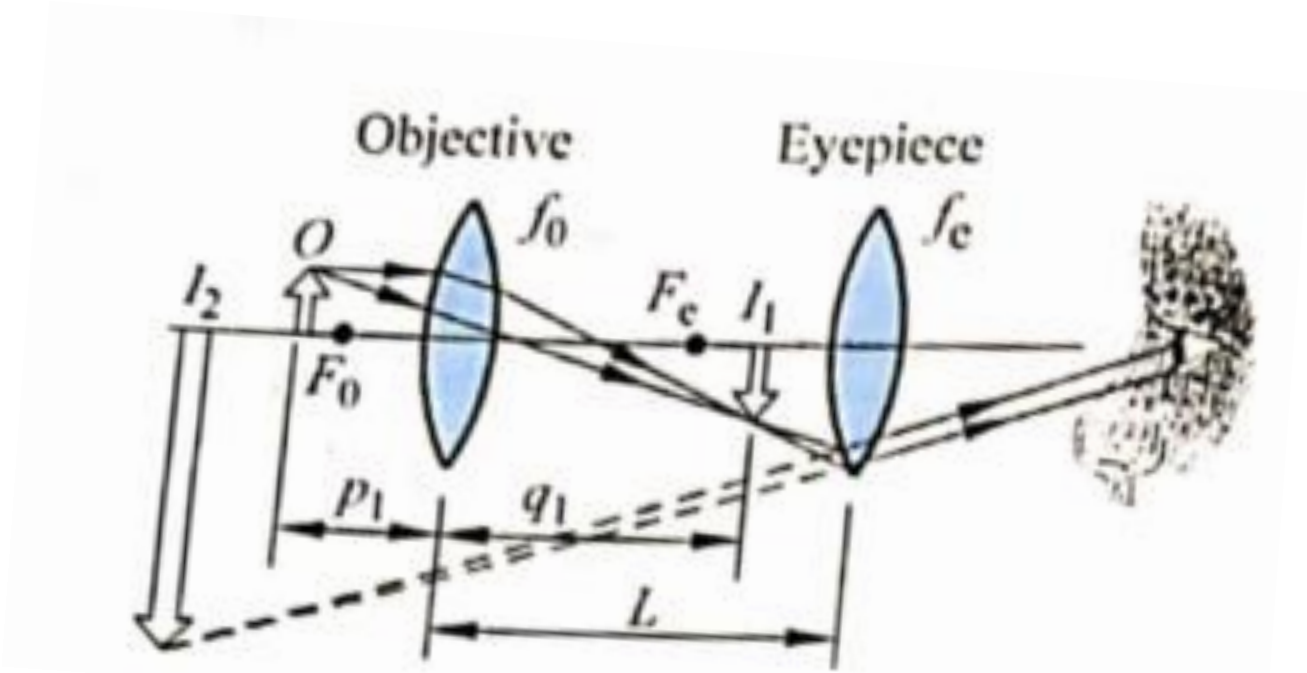


9.2.4 Telescope





9.2.4 Telescope



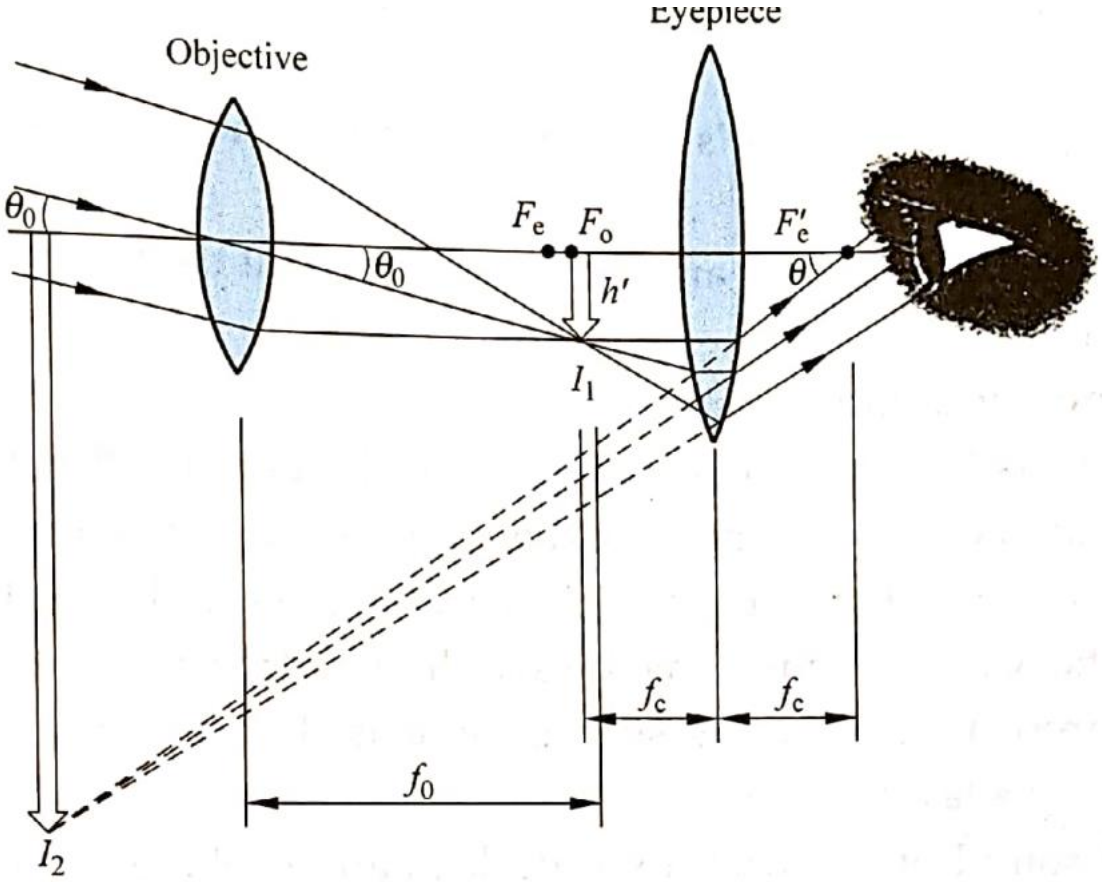
The magnification of a microscope

$$M \approx \frac{L}{f_0}$$

L: the distance between two lens

f_0: the focal length of objective lens

9.2.4 Telescope

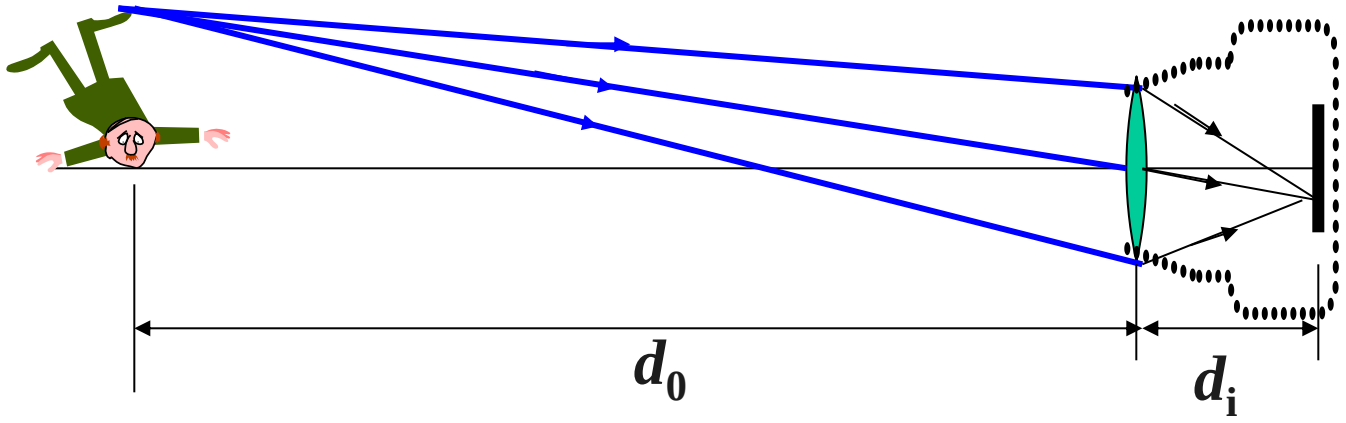


$$m = \frac{f_0}{f_e}$$

f_0: focal length of objective lens

f_e: focal length of eyepiece lens

9.2.4 Camera



9.2.5 Light in medium

Light speed in vacuum $c = 1/\sqrt{\varepsilon_0 \mu_0}$

Light speed in medium $u = 1/\sqrt{\varepsilon \mu}$

$$\frac{u}{c} = \frac{1}{n}$$

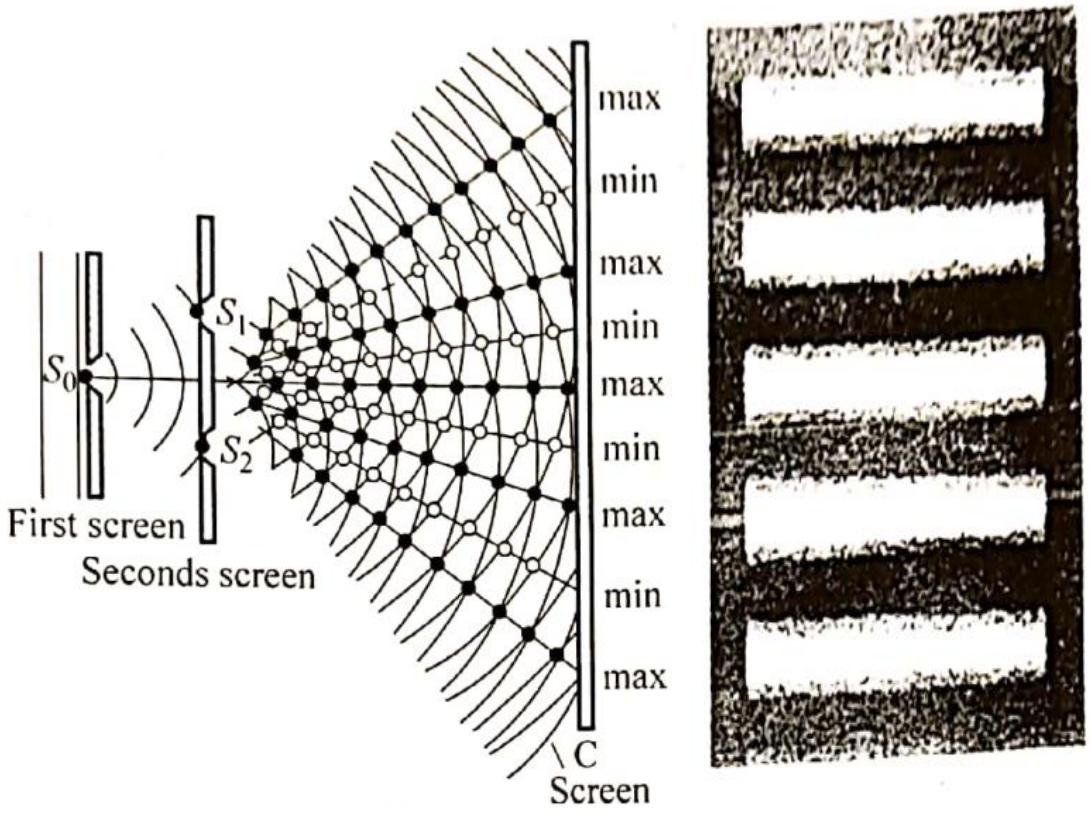
$$u = \lambda' \nu \qquad c = \lambda \nu$$

Wavelength $\lambda' = \frac{\lambda}{n}$

In vacuum

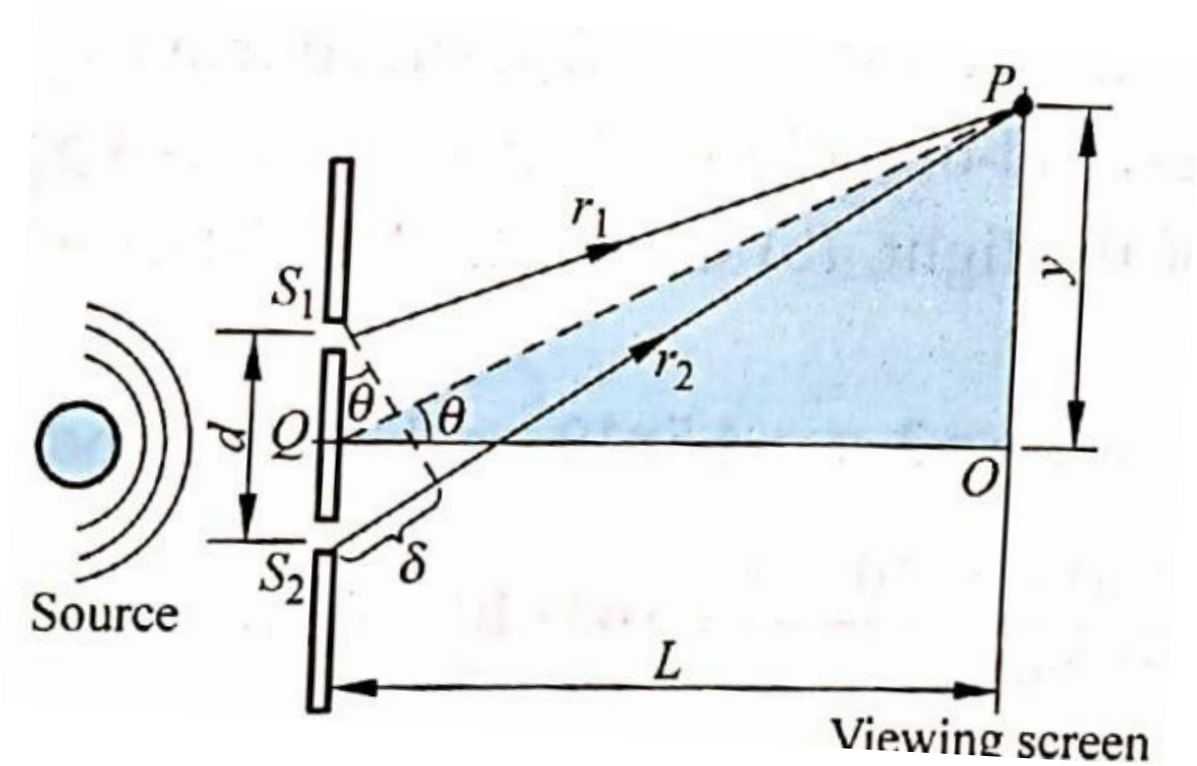
Refraction index

9.3 Interference: Young's double-slit experiment



(a) Schematic diagram of Young's double-slit experiment. The narrow slits act as Slits S1 and S2 behave as coherent sources that produce an interference pattern on screen drawing is not to scale. (b) The fringe pattern formed on screen C.

9.3.1 Young's Double-slit Experiment

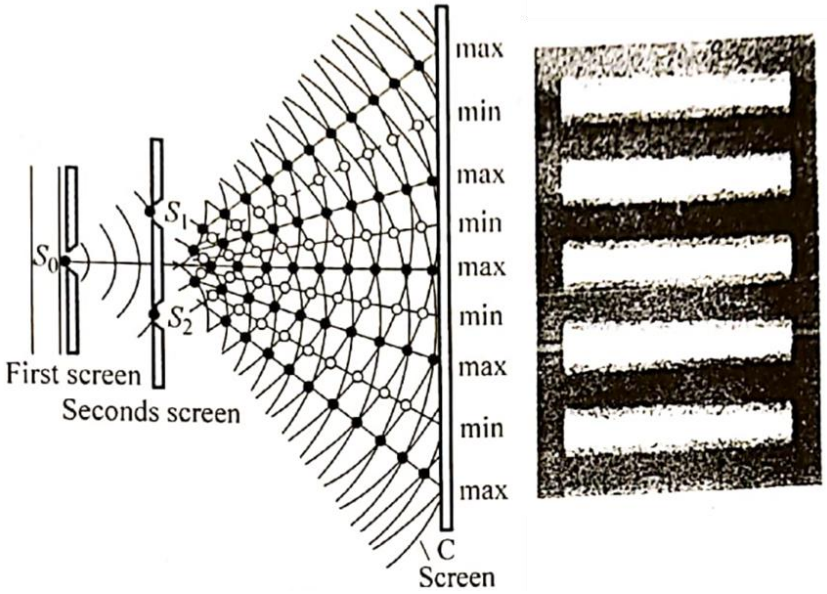


$$Light\ path\ difference$$

$$\delta = r_2 - r_1 = \mathrm{d}\sin\theta$$

d: distance between two slits

9.3.1 Young's Double-slit Experiment

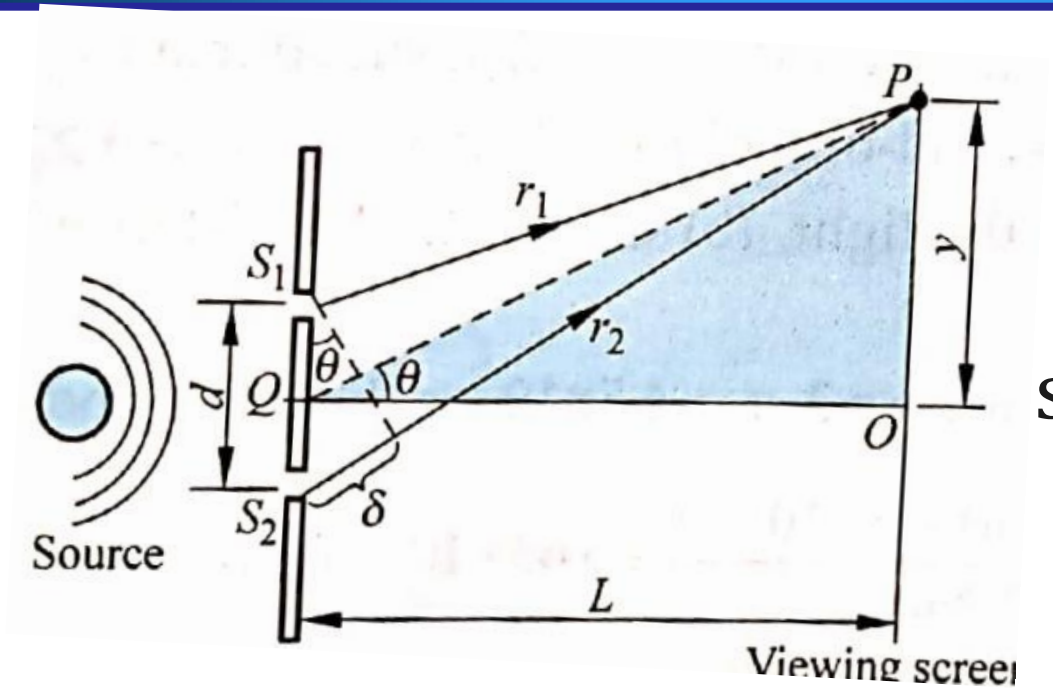


$$\delta = \mathrm{d}\sin\theta = \begin{cases} m\lambda & \text{Bright fringes} \\ (m+\frac{1}{2})\lambda & \text{Dark fringes} \end{cases}$$

m= 0, ±1, ± 2,……

d: distance between two slits
m: order number
λ: wavelength

9.3.1 Young's Double-slit Experiment



$$\sin\theta = tan\theta = \frac{y}{L}$$

$$y_{bright} = \frac{\lambda L}{d}\,\mathrm{m}$$

m= 0, ±1, ± 2,……

$$y_{dark} = \frac{\lambda L}{d}\left(\mathrm{m} + \frac{1}{2}\right)$$

9.3.1 Experiment

Example 1. A viewing screen is separated from a double-slit source by 1.2m.The distance between the two slits is 0.030mm. The second-order bright fringe (m = 2) is 4.5cm from the center line, (a) Determine the wavelength of the light, (b) Calculate the distance between adjacent bright fringes.

9.3.1 solution

Solution: (a)

$$y_{bright} = \frac{\lambda L}{d}\,\mathrm{m}$$

$$\lambda = \frac{dy_2}{mL} = \frac{(3.0*10^{-5}\ \mathrm{m})(4.5*10-2\mathrm{m})}{2*(1.2\ \mathrm{m})} = 5.63*10^{-7}\ \mathrm{m} = 563\ \mathrm{nm}$$

(b)

$$y_{m+1} - ym = \frac{\lambda L}{d}(m+1) - \frac{\lambda L}{d}m = \frac{\lambda L}{d} = 2.25\ \mathrm{cm}$$

9.3.1 Example

A light source emits visible light of two wavelengths: λ =430 nm, and λ'=510nm. The source is used in a double-slit interference experiment in which L = 1.5m and d=0.025mm. Find the separation between the third-order bright fringes.

9.3.1 Solution

Solution:

$$y_3 = \frac{\lambda L}{d}\,\mathrm{m} = \frac{\lambda L}{d}\,3$$

$$y'3 = \frac{\lambda' L}{d}\,\mathrm{m} = \frac{\lambda' L}{d}\,3$$

$$\Delta y = y_3 - y'_3 = \frac{(\lambda' - \lambda)\mathrm{L}}{\mathrm{d}}\,3 = \frac{1.5\ \mathrm{m} * (5.1 * 10^{-7} - 4.3 * 10^{-7})}{2.5 * 10^{-5}\ \mathrm{m}} * 3 = 1.44\ \mathrm{cm}$$

Homework

Page 218-219: Q10, Q12,

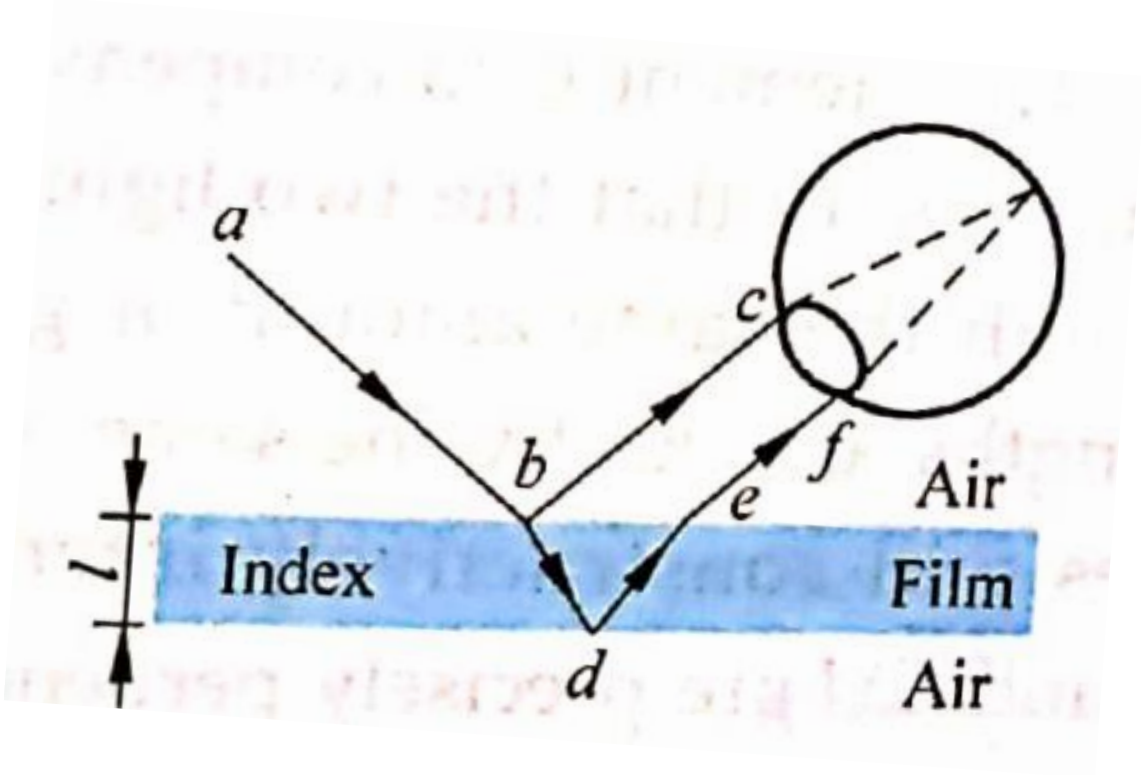
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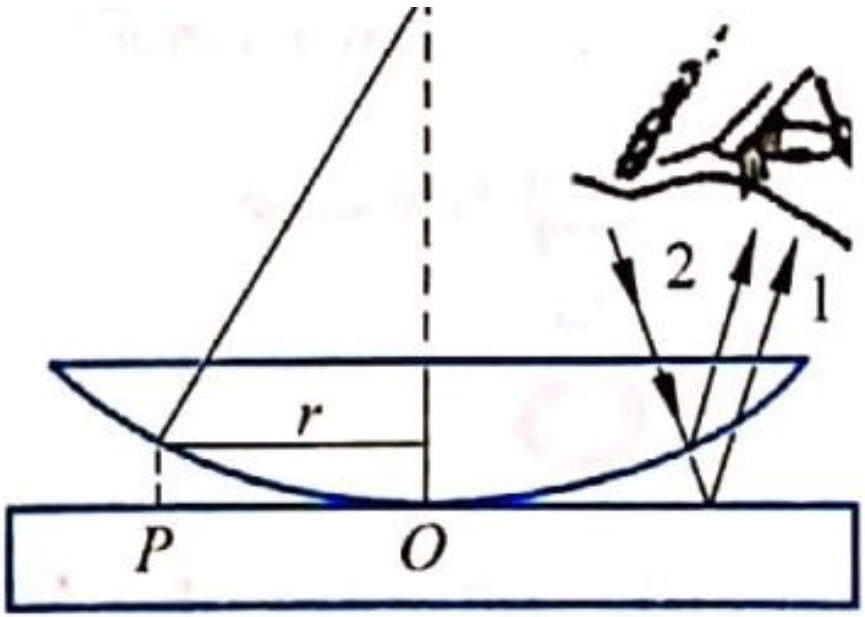
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9.3.2. Interference from Thin Films



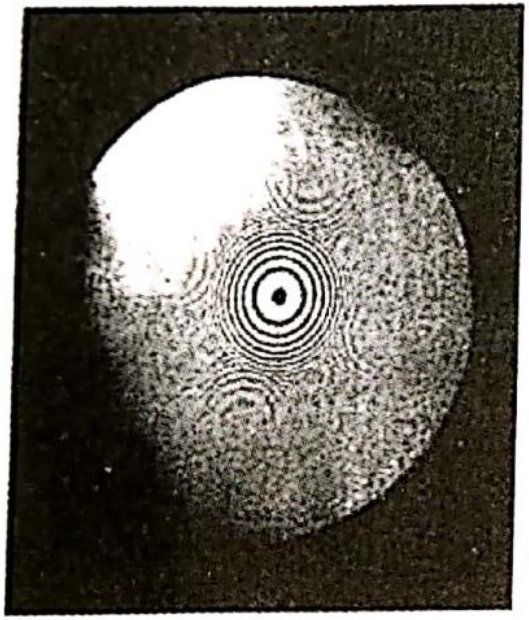
Interference between rays reflected from the upper and lower surfaces of a thin film.

9.3.2. Interference from Thin Films: Newton's rings



The geometry used to obtain the conditions for constructive or destructive interference in Newton's rings. The two rays moving toward the eye interfere after they reflect from different surfaces.

9.3.2. Interference from Thin Films

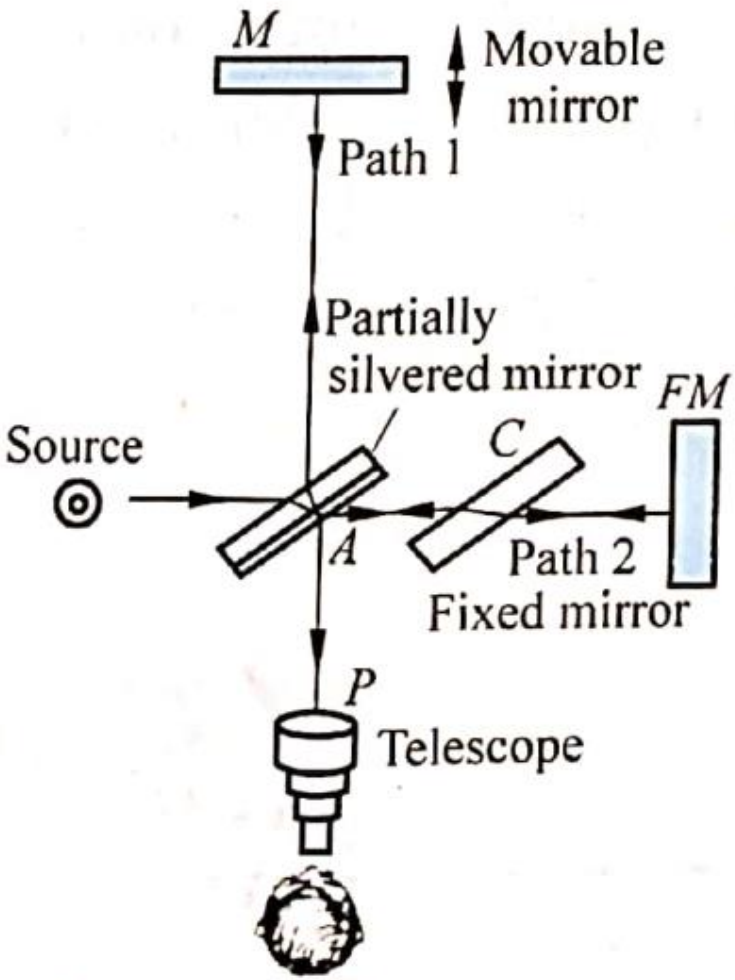


Newton’s rings formed by interference the air film between a convex and a plane surface



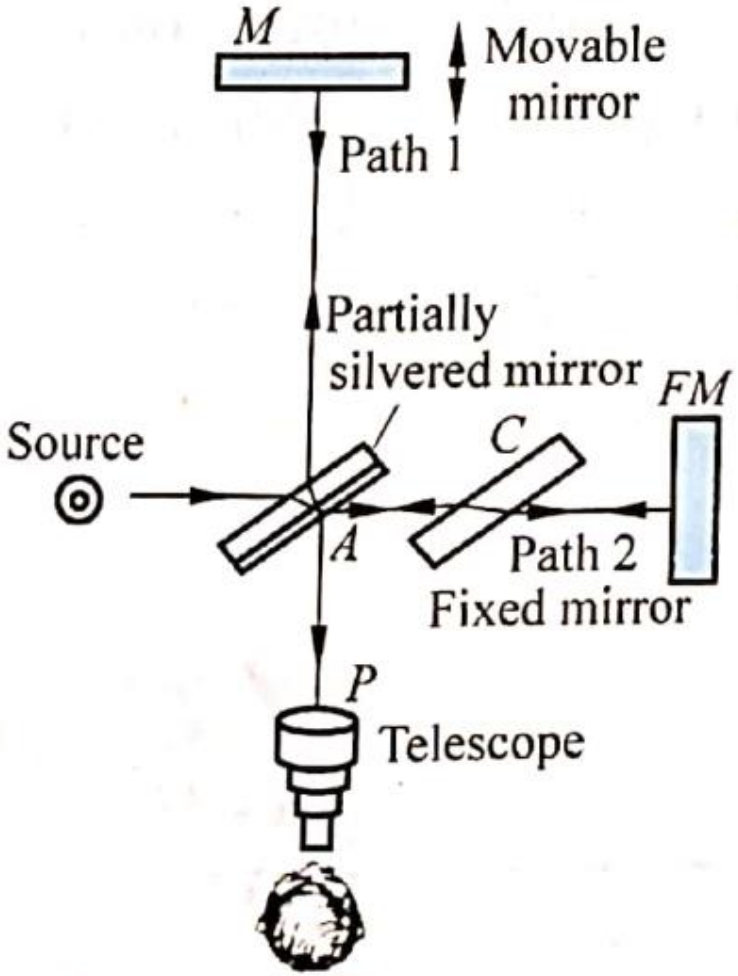
The asymmetrical interference in pattern of Newton’s rings indicates imperfections in the lens

9.3.2. Michelson interferometer



Schematic diagram of a Michelson interferometer. Light is split by the partially silvered mirror. The resulting light travels two different paths before it returns to and interferes at point *A* and is subsequently observed through the telescope.

9.3.2. Michelson interferometer



Unknown wavelength can be determined by measuring very accurately the movement of the movable mirror and counting the number of maxima that pass across the telescope eyepiece.

$$N = \frac{2\Delta L}{\lambda}$$

N: the number of the maxima pass

ΔL: the distance the mirror moved

λ: wavelength

$$\lambda = \frac{2\Delta L}{N}$$

9.3.2. Example

Light from a helium-neon laser is beamed into a Michelson interferometer. The movable mirror is displaced 0.567mm, causing the interference pattern to reproduce itself 1800 times.

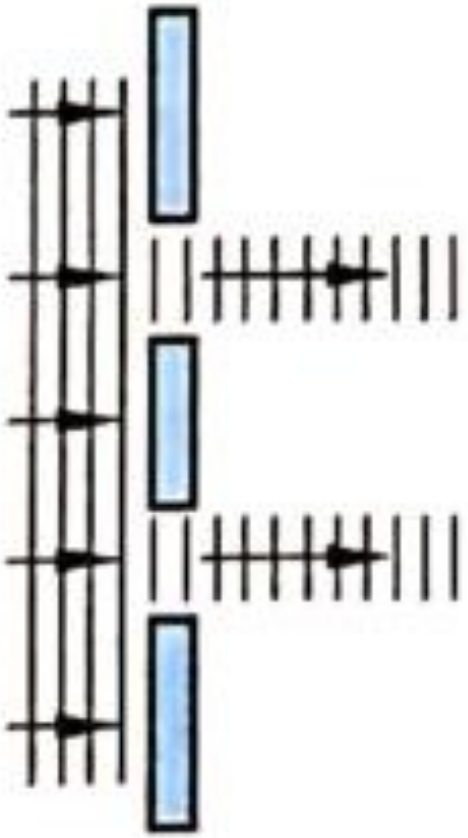
Find the wavelength of helium-neon laser light.

9.3.2. Solution

Solution: The wavelength follows:

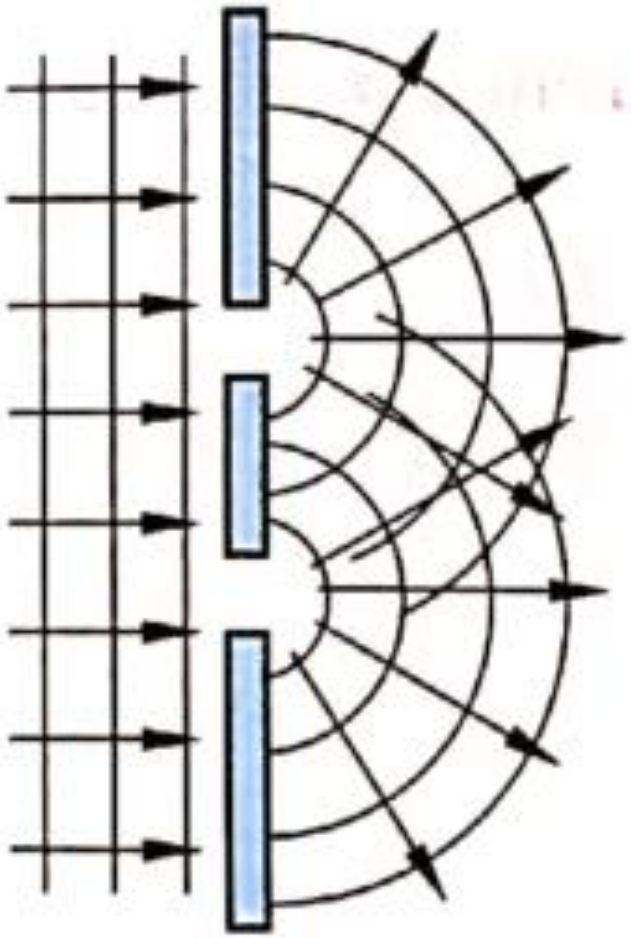
$$\lambda = \frac{2\Delta L}{N} = \frac{2(0.567 * 10^{-3}\text{m})}{1800} = 630 \text{ nm}$$

9.4 Diffraction



Common sense: The light will go straight when the size of opening/objects is much larger than that of light wavelength

9.4 Diffraction

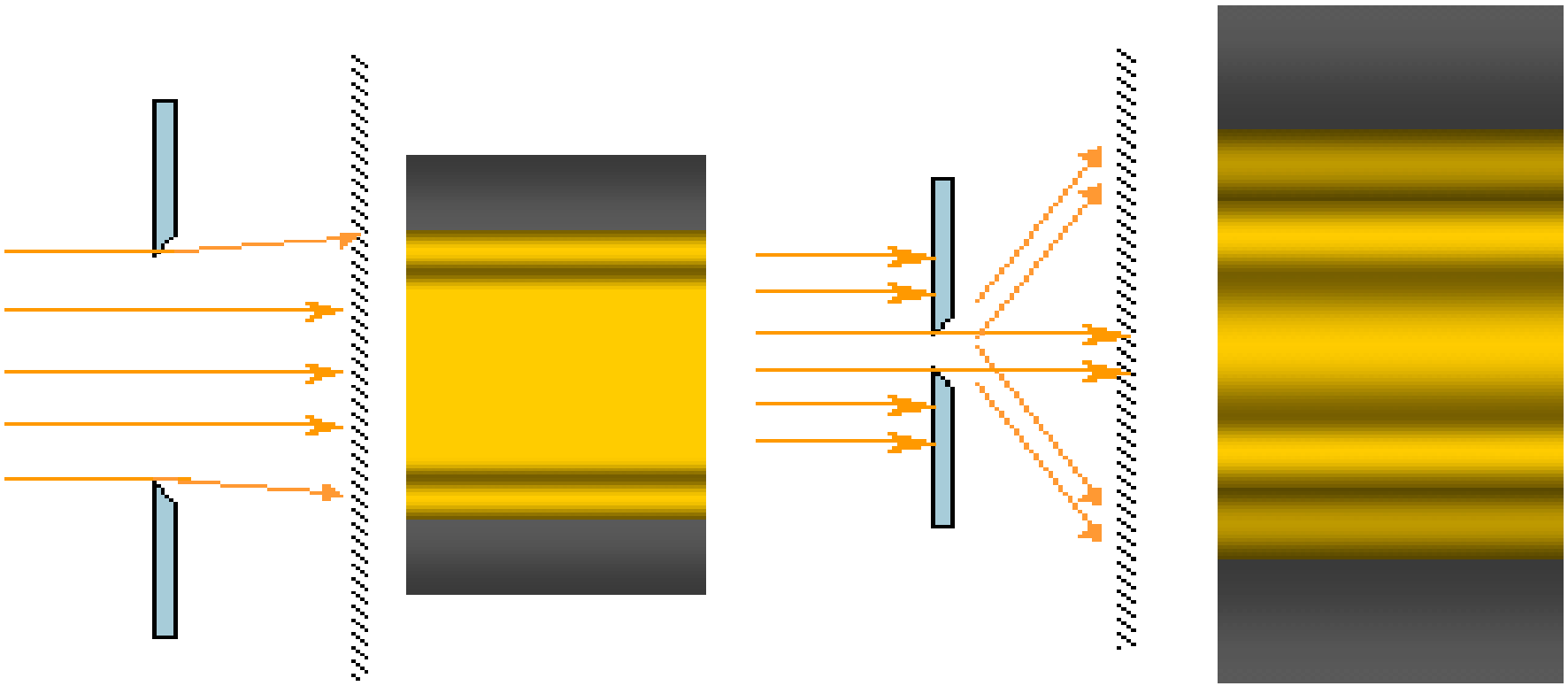


Conditions: the size of comers/opening/objects is comparable with that of light wavelength

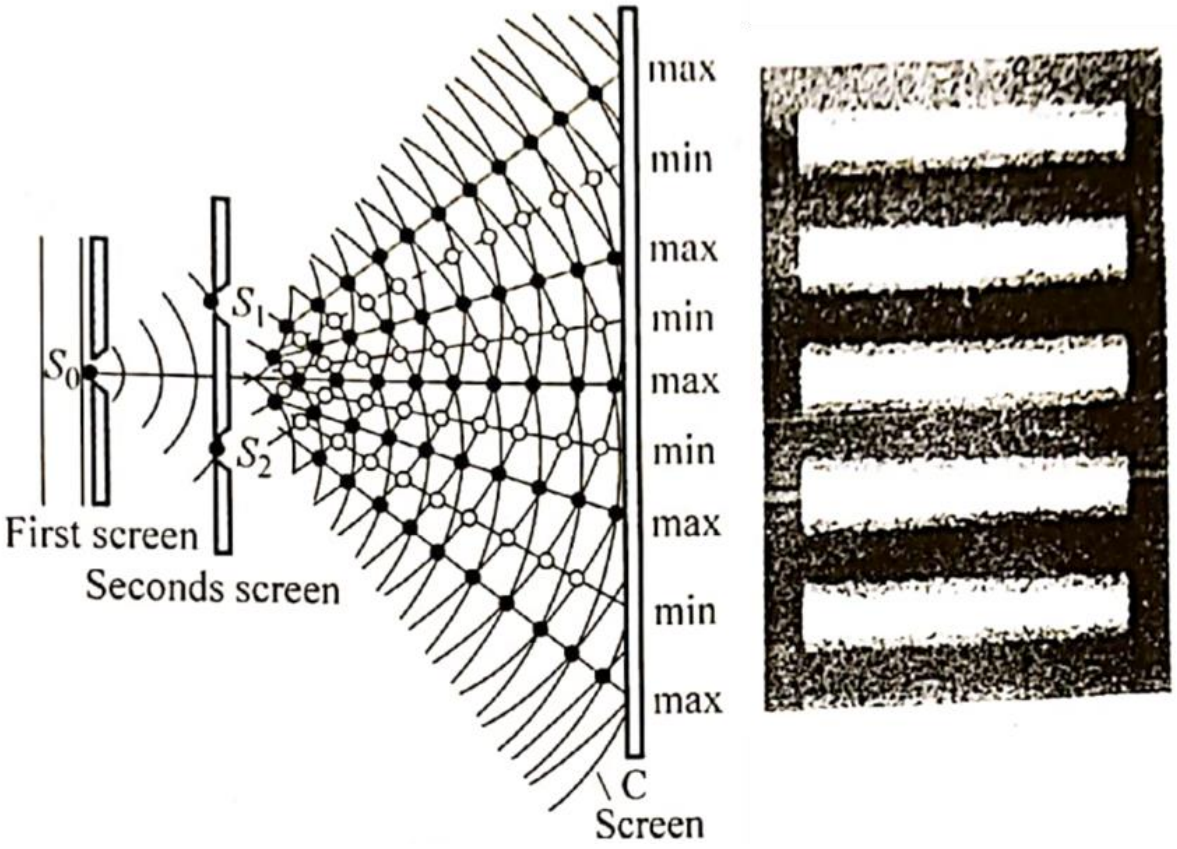
Diffraction: When a light passing through a small aperture, an interference pattern is observed rather than a sharp spot of light, showing that light spreads beyond the aperture into regions where a shadow would be expected if light traveled in straight lines.

The behavior of light waves is similar with that of water wave, has the property of being able to bend around comers.

9.4 Diffraction



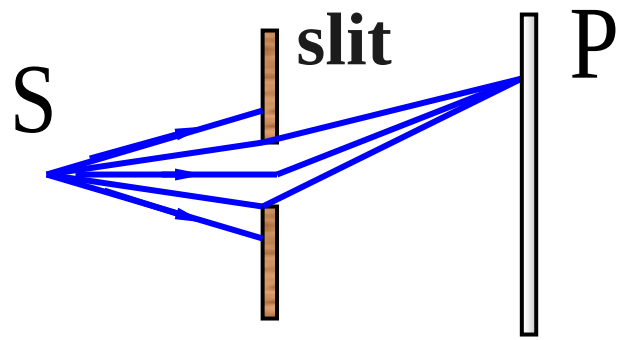
9.4 Diffraction



Young’s double-slit inference is due to the light diffraction

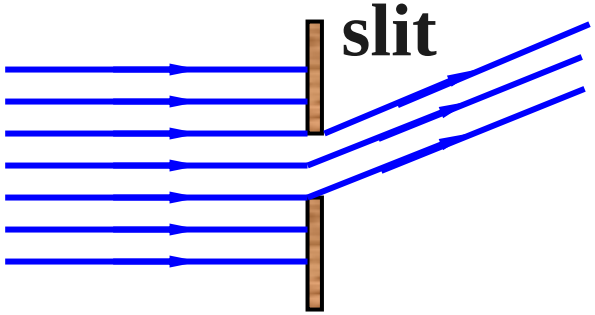
9.4 Diffraction

Fresnel diffraction



Source, slit and panel: finite distance

Fraunhofer diffraction

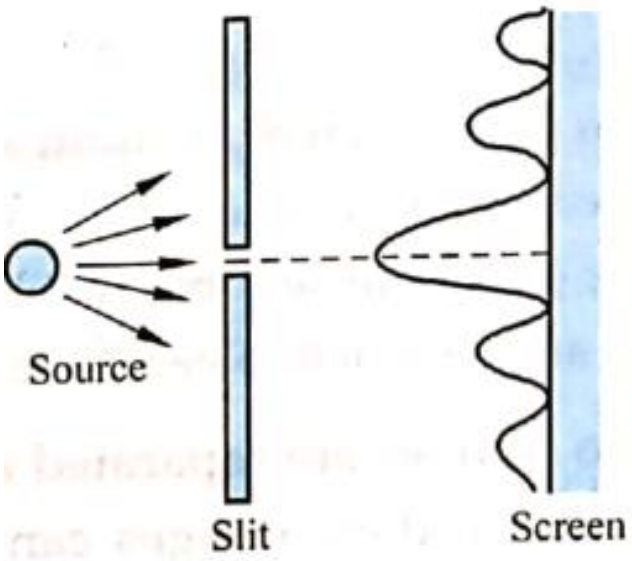


Source, slit and panel: infinite distance

Light: parallel line

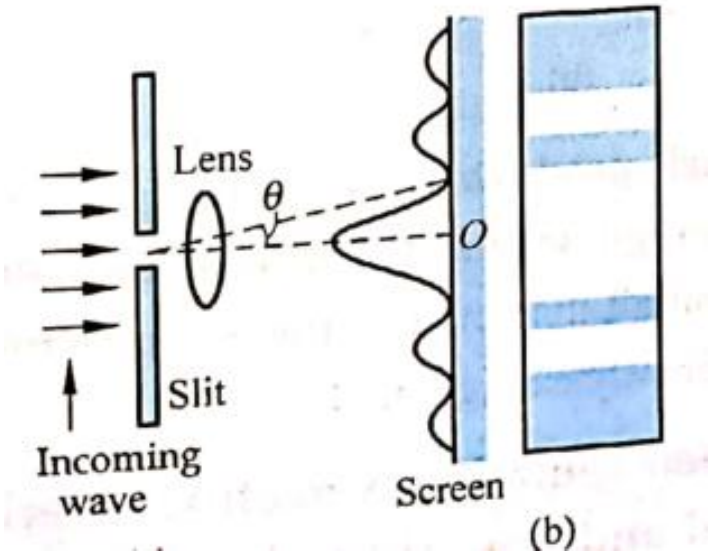
9.4 Diffraction

Fresnel diffraction



Source, slit and panel: finite distance

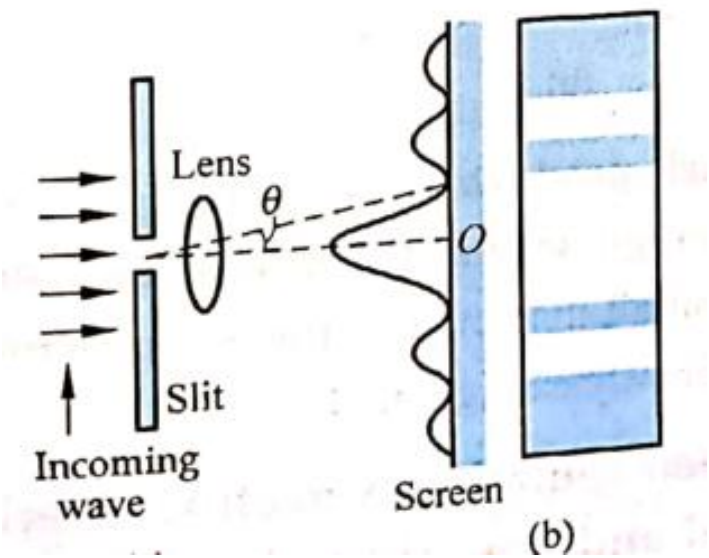
Fraunhofer diffraction



Source, slit and panel: infinite distance
Light: parallel line

9.4 Diffraction： Fraunhofer diffraction

The minimum intensity (dark position) of the diffraction pattern on the screen occurs



$$a\sin\theta = m\lambda$$

$m = \pm 1, \pm 2, \ldots\ldots$

a: the width of the slit

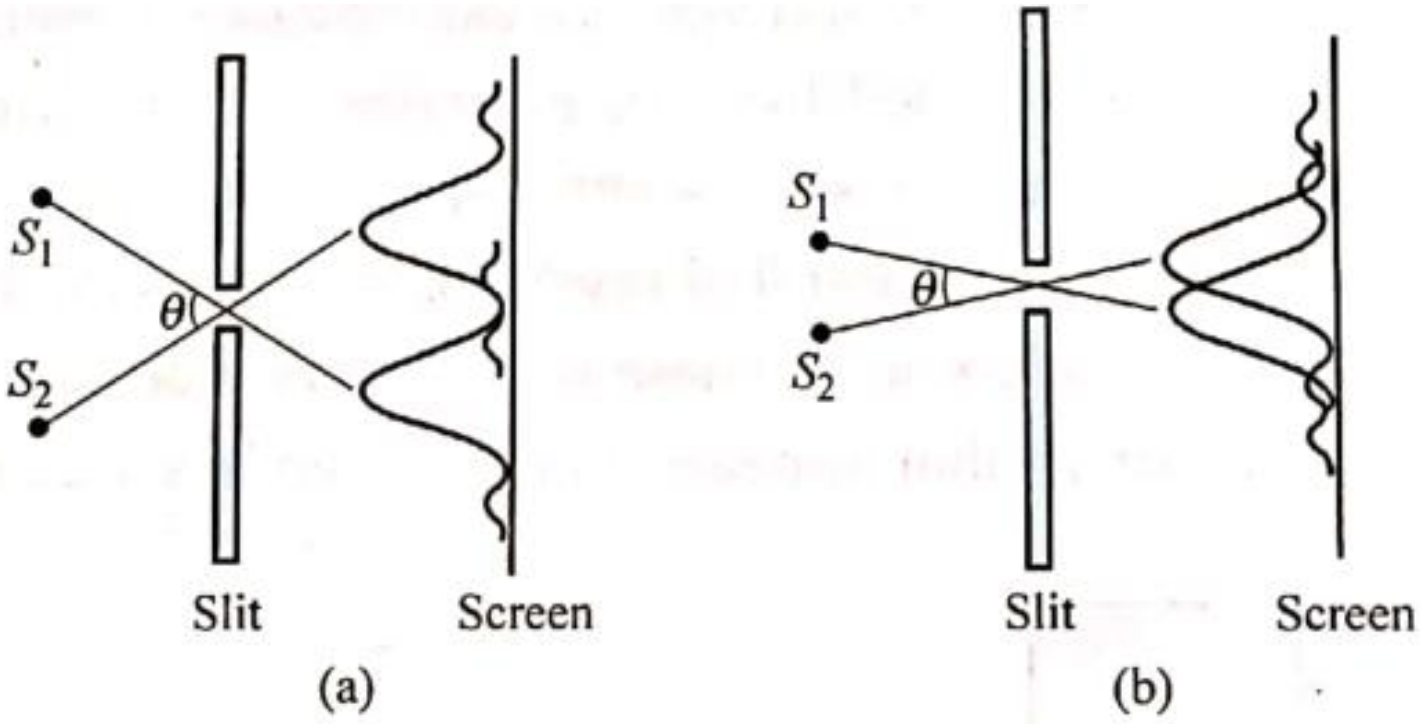
For m=1: $a\sin\theta = \lambda$

the first minimum intensity (dark)

When a ↓, θ↑

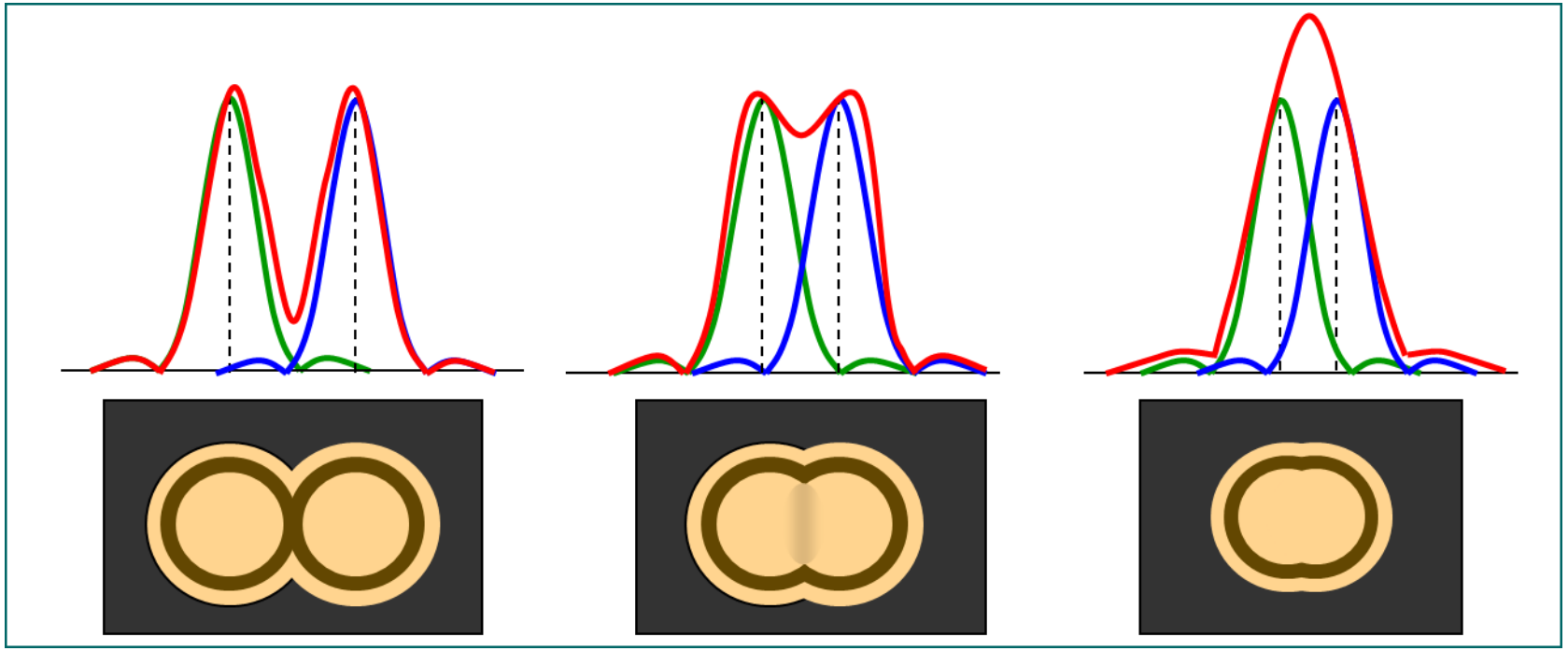
When a= λ, θ=90°, the central bright fills the full screen.

9.4.2 The Resolution of optical instruments



Two point sources far from a small aperture each produce a diffraction pattern (a) The angle subtended by the sources at the aperture is large enough so that the diffraction patterns are distinguishable; (b) The angle subtended by the sources is so small that their diffraction patterns overlap and the images are not well resolved (Note that the angles are greatly exaggerated).

9.4.2 Rayleigh's criterion: judge the two images resolved



Two images just resolved (judge by Rayleigh's criteria: when the central maximum of one image falls on the first minimum of another image.

9.4.2 Rayleigh's criterion: judge the two images resolved

Based on Rayleigh's criterion: the minimum angular separation, θ_{min} , so that their images can be just resolved.

$$\theta_{min} = 1.22\frac{\lambda}{D}$$

D: the diameter of the aperture

The smaller, the greater to be the resolving power

9.4.2 Example

Light of wavelength 589nm (sodium yellow light) is used to view an object under a microscope. If the aperture of the objective has a diameter of 0.900cm.

(a) Find the limiting angle of resolution,

(b) Using visible light of any wavelength you desire, what is the maximum limit of resolution for this microscope?

9.4.2 Example

Solution:

(a) The limiting angle of resolution to be:

$$\theta_{min} = 1.22\frac{\lambda}{D} = 1.22\left(\frac{589*10^{-9}\text{m}}{0.9*10^{-2}\text{m}}\right) = 7.89*10^{-5}\text{ rad}$$

(b) To obtain the smallest angle corresponding to the maximum limit of resolution, we have to use the shortest wavelength available in the visible spectrum. Violet light (400nm) gives us a limiting angle of resolution of

$$\theta_{min} = 1.22\frac{\lambda}{D} = 1.22\left(\frac{400*10^{-9}\text{m}}{0.9*10^{-2}\text{m}}\right) = 5.42*10^{-5}\text{ rad}$$

Homework

Page 218-219: Q13, Q14, Q16,

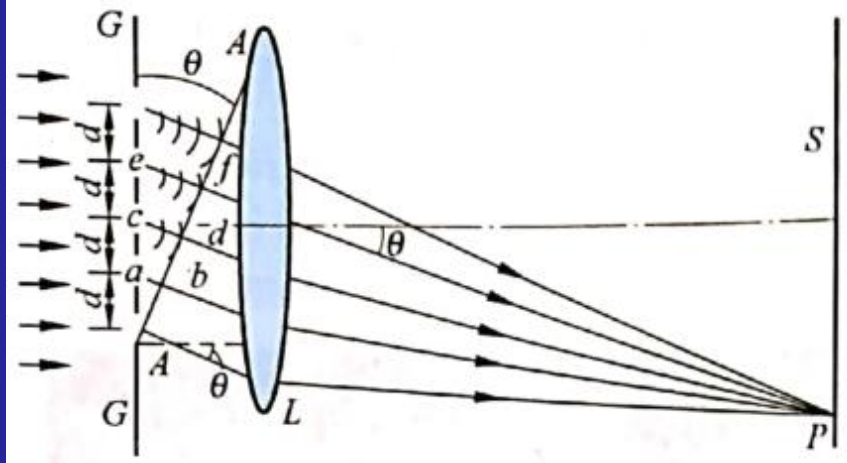
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for example: 2023800901XX + week 11 + Alfa Robot

pdf file, send to the cloud)

9.4.3 The diffraction grating



The diffraction grating: a useful device tor analyzing light sources, consists of a large number of equally spaced parallel slits.

Gratings with many lines very close to each other can have very small slit spacings. For example, a grating ruled with 5000 lines cm^{-1}, the slit spacing d= $2 * 10^{-4}$ cm=2 μm

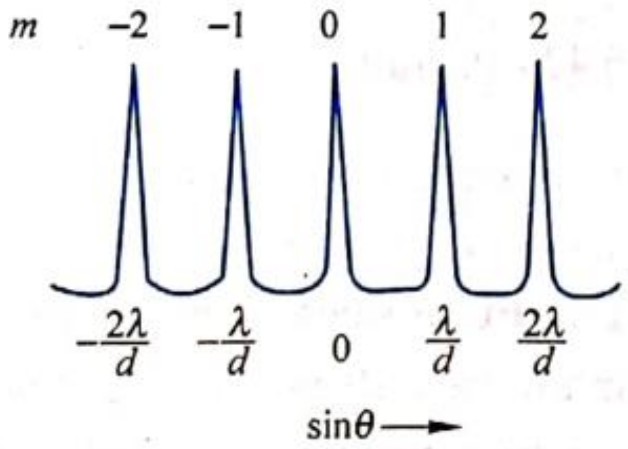
$$d \sin\theta = m\lambda$$

d: *slit separation*

m= 0, ±1, ± 2,……

θ=0, m=0, zeroth-order maximum.

9.4.3 The diffraction grating



Intensity versus sinθ for a diffraction grating. The zeroth-, first- and second-order maxima are shown.

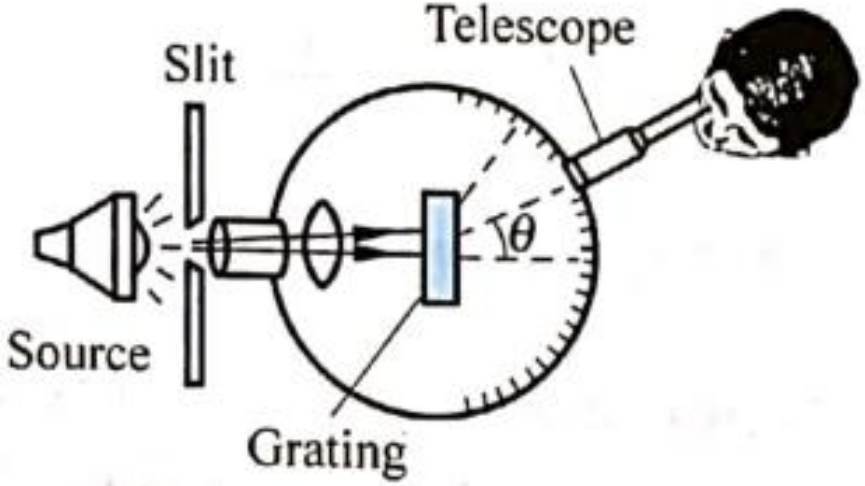


Diagram of a diffraction grating spectrometer.
The collimated beam incident on the grating is diffracted into the various orders at the angles θ that satisfy the equation $d\sin\theta=m\lambda$

9.4.3 Example

Monochromatic light from a helium-neon laser (λ=632.8 nm) is incident normally on a diffraction grating with 6000 lines cm^{-1}. Find the angles at which the first-order, second-order and third-order maximum can be observed.

Solution:

The slit separation d=(1/6000) cm= 1.667 * 10^{-4} cm=1667 nm

For the first-order maximum: m=1,

$$d \sin\theta = m\lambda$$

$$\sin\theta = \frac{\lambda}{d} = \frac{632.8nm}{1667nm} = 0.3796 \qquad \theta = 22.31^{o}$$

9.4.3 Example

For the second-order maximum: m=2,

$$\sin\theta = \frac{2\lambda}{d} = \frac{2 * 632.8nm}{1667nm} = 0.7592$$

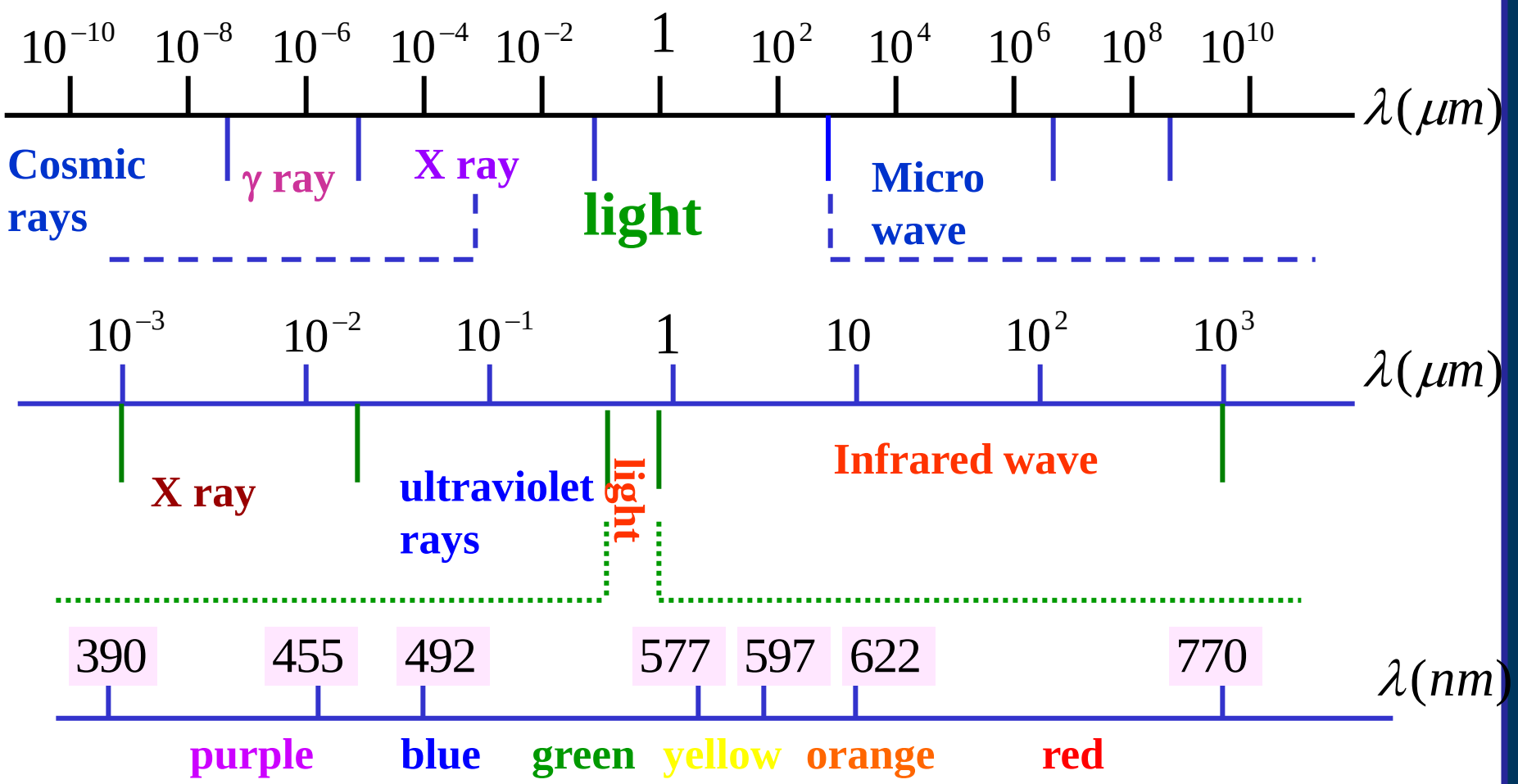
$$\theta = 49.41^{o}$$

For the third-order maximum: m=3,

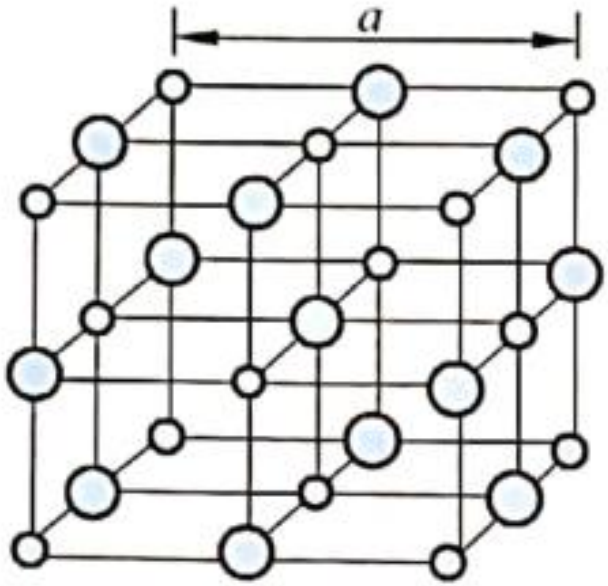
$$\sin\theta = \frac{3\lambda}{d} = \frac{3 * 632.8nm}{1667nm} = 1.139$$

$Sin\theta$<=1, therefore, there is no third-order maximum.

9.4.4 Diffraction X-rays by Crystals

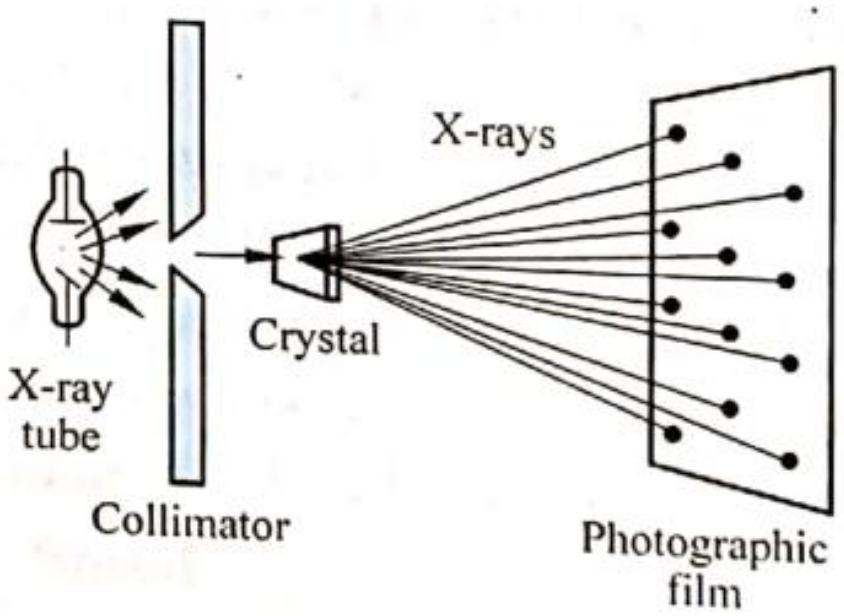


9.4.4 X-rays Diffraction



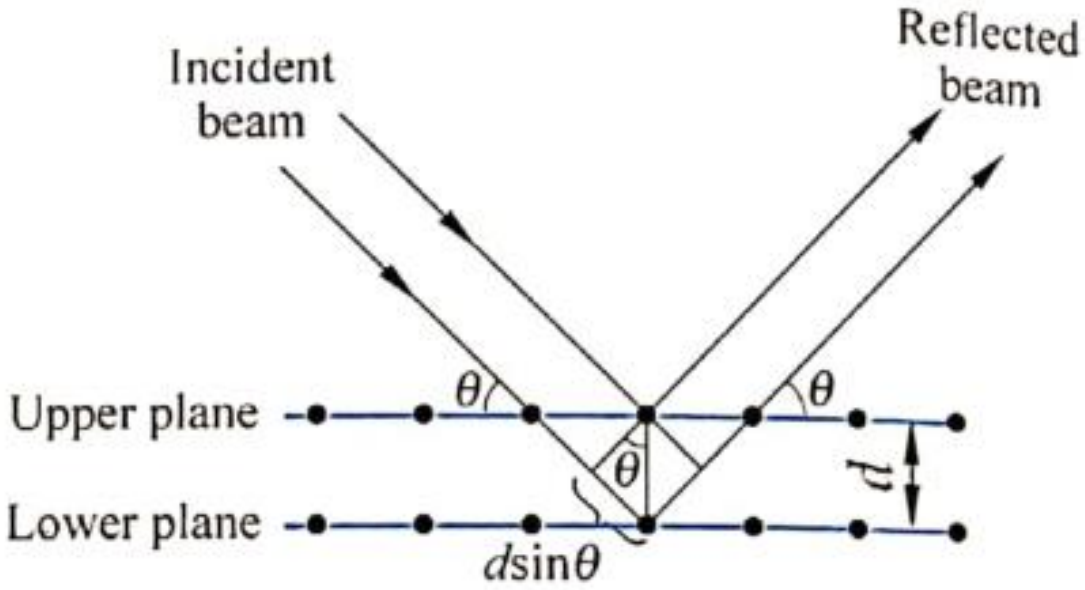
A schematic diagram of a crystal (NaCl). The length of the cube edge is a=0.563 nm

A basic XRD setup



Cu target X-ray: 0.15418nm

9.4.4 X-rays Diffraction: determine materials



Bragg's law

$$2d\sin\theta = m\lambda \qquad m = 1,2,3\ldots\ldots$$

9.4 Diffraction

An X-ray tube produces a continuous distribution of wavelength. If these wavelengths are scattered from a particular set of parallel planes of rock salt (NaCl) with a spacing distance d = 0.282nm, what wavelengths will appear in the first and second orders at Θ = 25° ?

Solution:

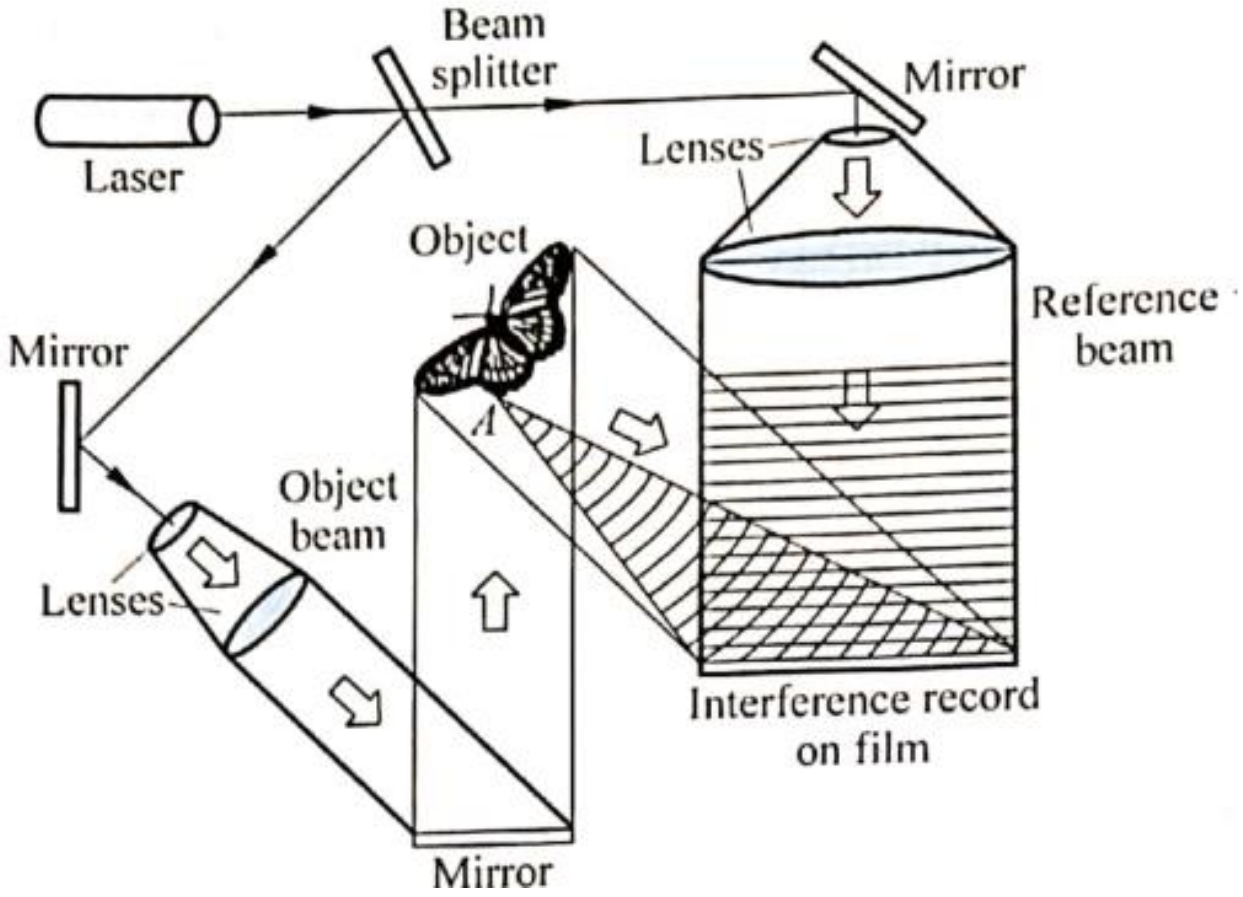
$$2d\sin\theta = m\lambda$$

$$\lambda = \frac{2d\sin\theta}{m} = \frac{2(0.282\text{nm})(\sin 25^\circ)}{m} = 0.238\text{nm/m}$$

The first-order: $\lambda = 0.238\text{nm}$

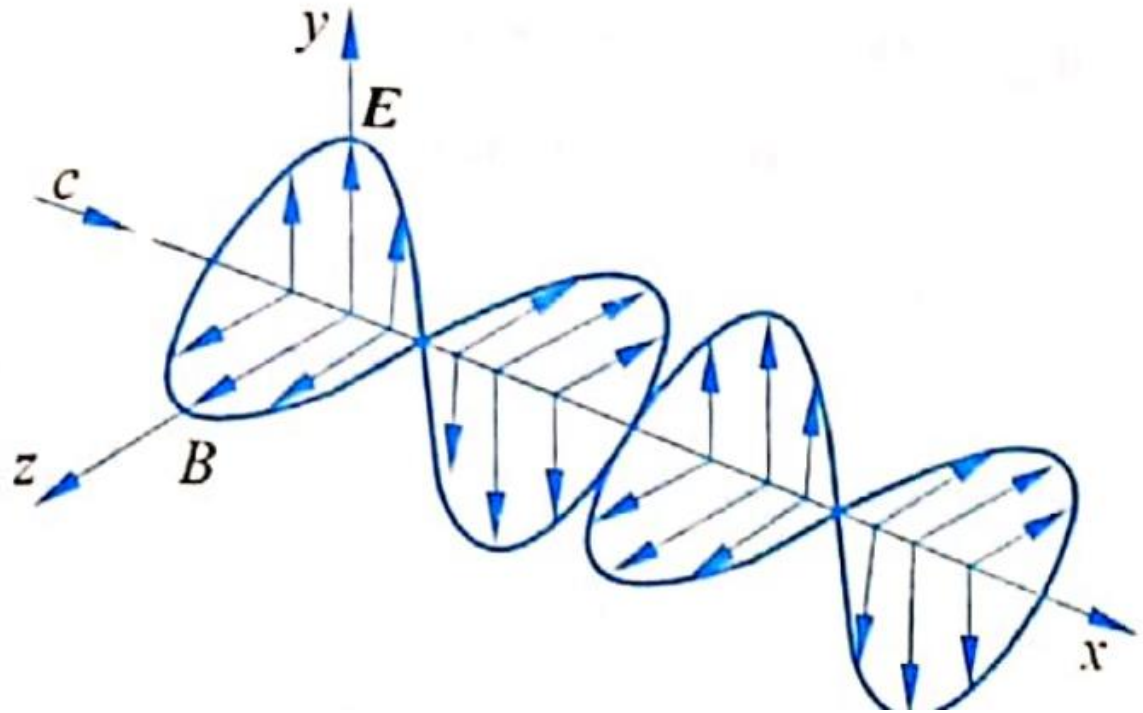
The second-order: $\lambda = 0.119\text{nm}$

9.4 Holography: record the 3D object



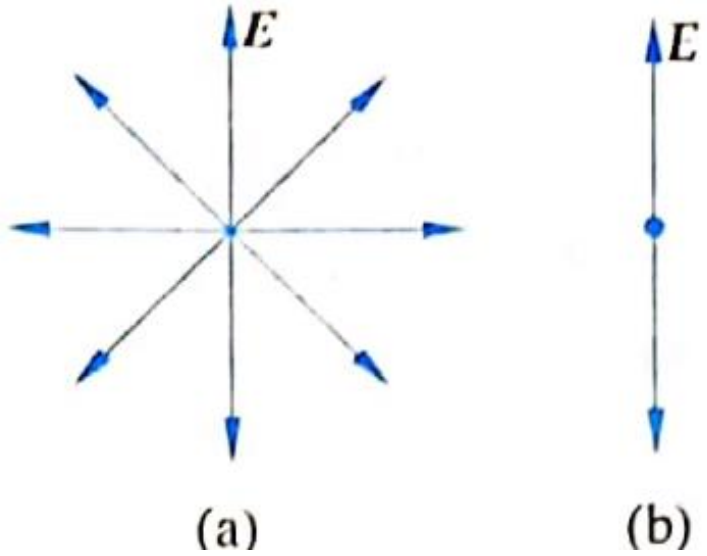
Schematic diagram of how laser light is split to form object and reference beams. The object beam reflects from the object. And the interference of the light is stored on a hologram

9.5 Polarization



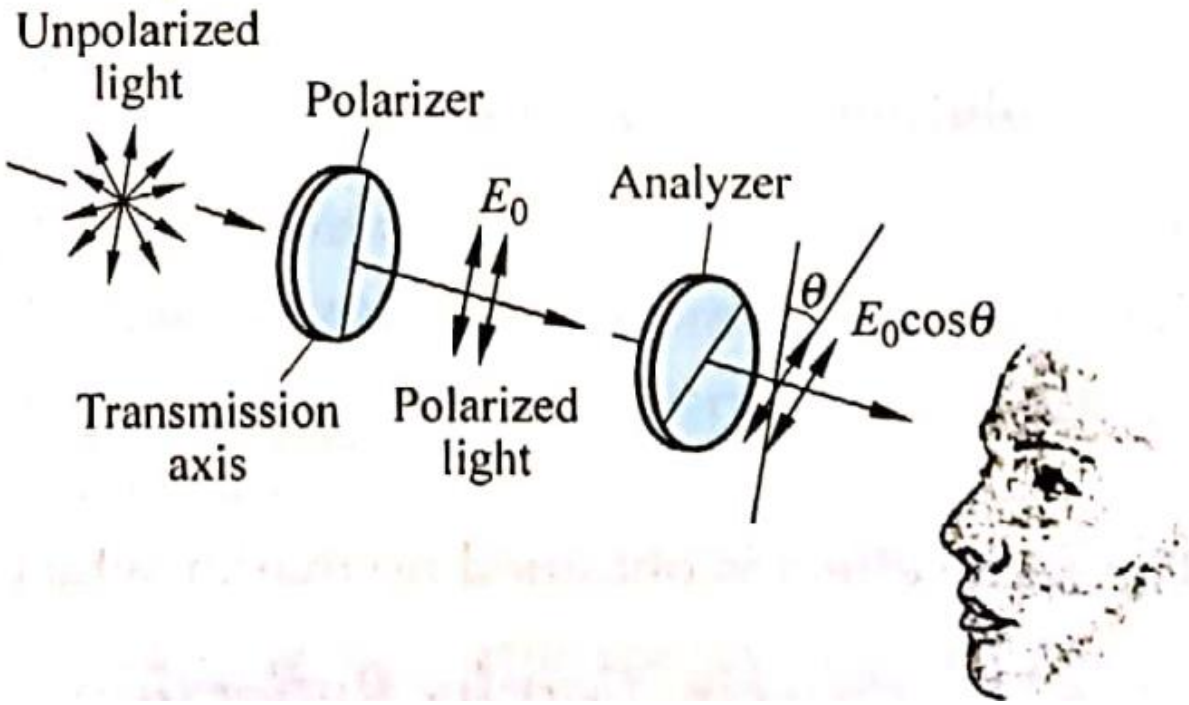
Schematic diagram of an electromagnetic wave propagating in the a: direction. The electric field vector *E* vibrates in the *xy* plane, and the magnetic field vector *B* vibrates in the *xz* plane.

9.5 Polarization



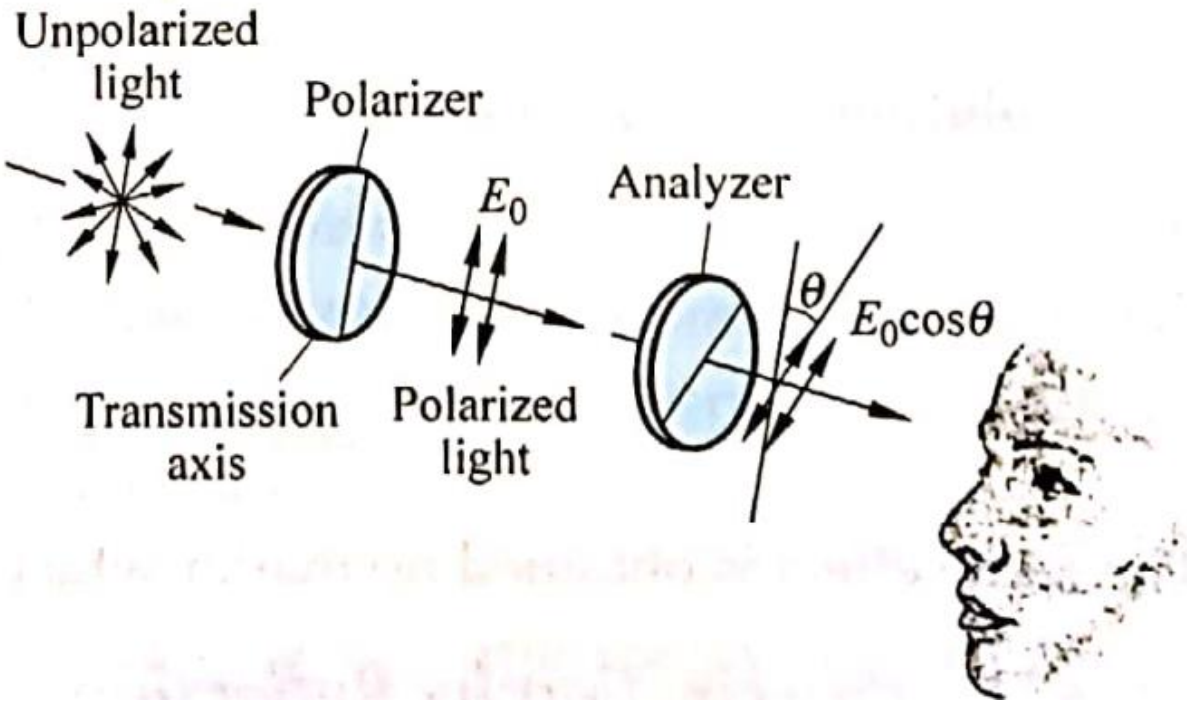
An unpolarized light beam viewed along the direction of propagation (perpendicular to the page). The transverse electric field vector can vibrate in any direction with equal probability; (b) A linearly polarized light beam with the electric field vector vibrating in the vertical direction.

9.5 Polarization: Malus' law



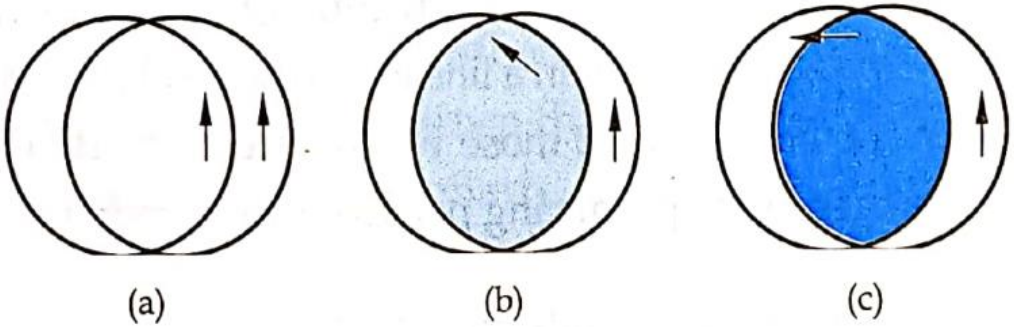
Two polarizing sheets whose transmission axes make an angle θ with each other. Only a fraction of the polarized light incident on the analyzer is transmitted.

9.5 Polarization: Malus' law



Malus' law $I = I_0 cos^2\theta$

9.5 Polarization: Malus' law



The intensity of light transmitted through two polarizers depends on the relative orientation of their transmission axes

(The transmitted light has maximum intensity when the transmission axes are aligned with each other;

(The transmitted light intensity diminishes when the transmission axes are at an angle of 45° with each other; (c) The transmitted light intensity is a minimum when the transmission axes are at right angles to each other.

$$I = I_0 cos^2\theta$$

9.5 Polarization

Two polarizing sheets have their polarizing directions parallel so that the intensity I_0 of the transmitted light is a maximum. Through what angle must either sheet be turned if the intensity is to drop by one-half?

Solution:

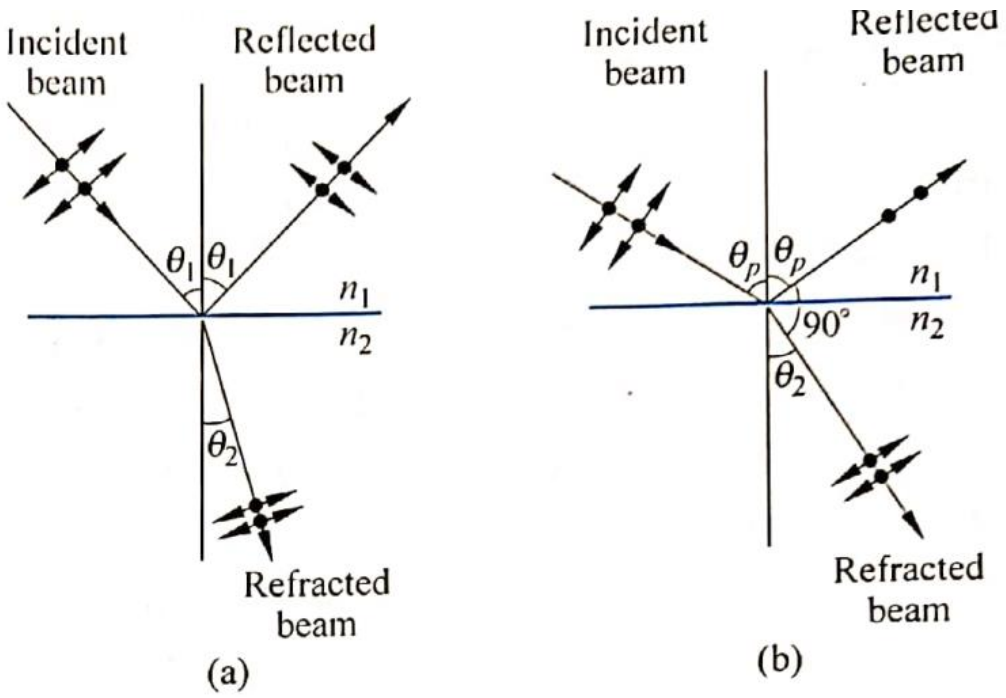
$$I = I_0 cos^2\theta$$

$$\frac{1}{2} I_0 = I_0 cos^2\theta$$

$$\theta = \pm 45^o \; and \pm 135^o$$

The same effect is obtained no matter which sheet is rotated or in which directions.

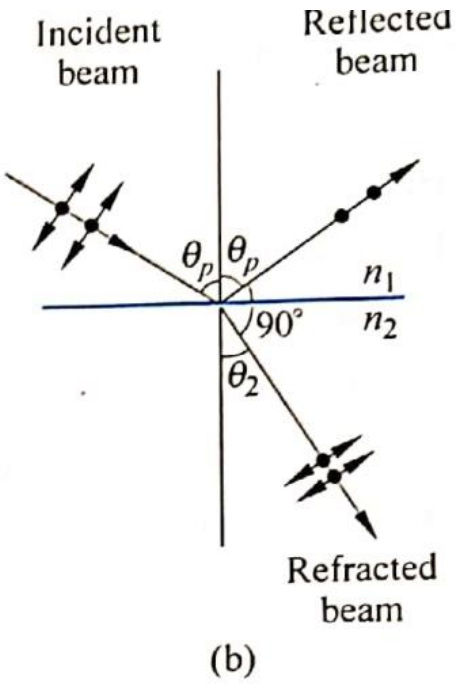
9.5 Polarization: Brewster's Law



When unpolarized light is incident on a reflecting surface, the reflected and refracted beams are partially polarized; (b) The reflected beam is completely polarized when the angle of incidence equals the polarizing angle,

9.5 Polarization: Brewster's Law

A reflected wave or light will be fully polarized with its electric field *E* vector perpendicular to the plane of the incidence or in other words, parallel to the boundary surface, if it strikes a boundary at the Brewster's angle θ_P



Brewster's Law

$$tan\theta_p = n_2/n_1$$

Questions

1. Sunlight or starlight passing through the earth's atmosphere is always bent toward the vertical. Why? Does this mean that a star isn't really where it appears to be?

2. Why does a plane mirror reverse left and right but not top and bottom?

3. For a plane mirror, what is the focal length? The magnification?

Questions

4. How do your eyes adjust for seeing objects at different distances from you?

5. Can a virtual image be photographed by exposing a film at the location of the image? Explain.

6. A light travels from one medium to another, does the wavelength of the light change? Doe the frequency change? Does the speed change? Explain.

7. Describe an experiment in which internal reflection is used to determine the index of refraction of a medium.

8. Consider the image formed by a thin converging lens. Under what conditions is the image (a) inverted, (b) upright, © real, (d) virtual, (e) larger than the object, and (f) smaller than the object.

Questions

9. What is the necessary condition on the path-length-difference between two waves that interfere (a) constructively and (b)destructively?

10. If Young's double-slit experiment were performed under water, how would the observed interference pattern be affected?

11. In Young’s double-slit experiment, why do we use monochromatic light? If white light is used, how would the pattern be affected?

12. Why is diffraction grating better than a two-slit setup for measuring wavelength of light?

Questions

13. Is it possible to obtain better resolution with a microscope with blue light than with red light? Why or why not?

14. The spreading of light due to diffraction in an optical instrument is greater when the instrument uses red as opposed to blue light. Why?

15.Is polarization or interference a better test for identifying waves? Do they give the same information?

Home work

Page 218-219: Q19, Q21, Q22

Homework file name:

Roll number+week 12 + name,

for example: 2023800901XX + week 12 + Alfa Robot

pdf file, send to the cloud)

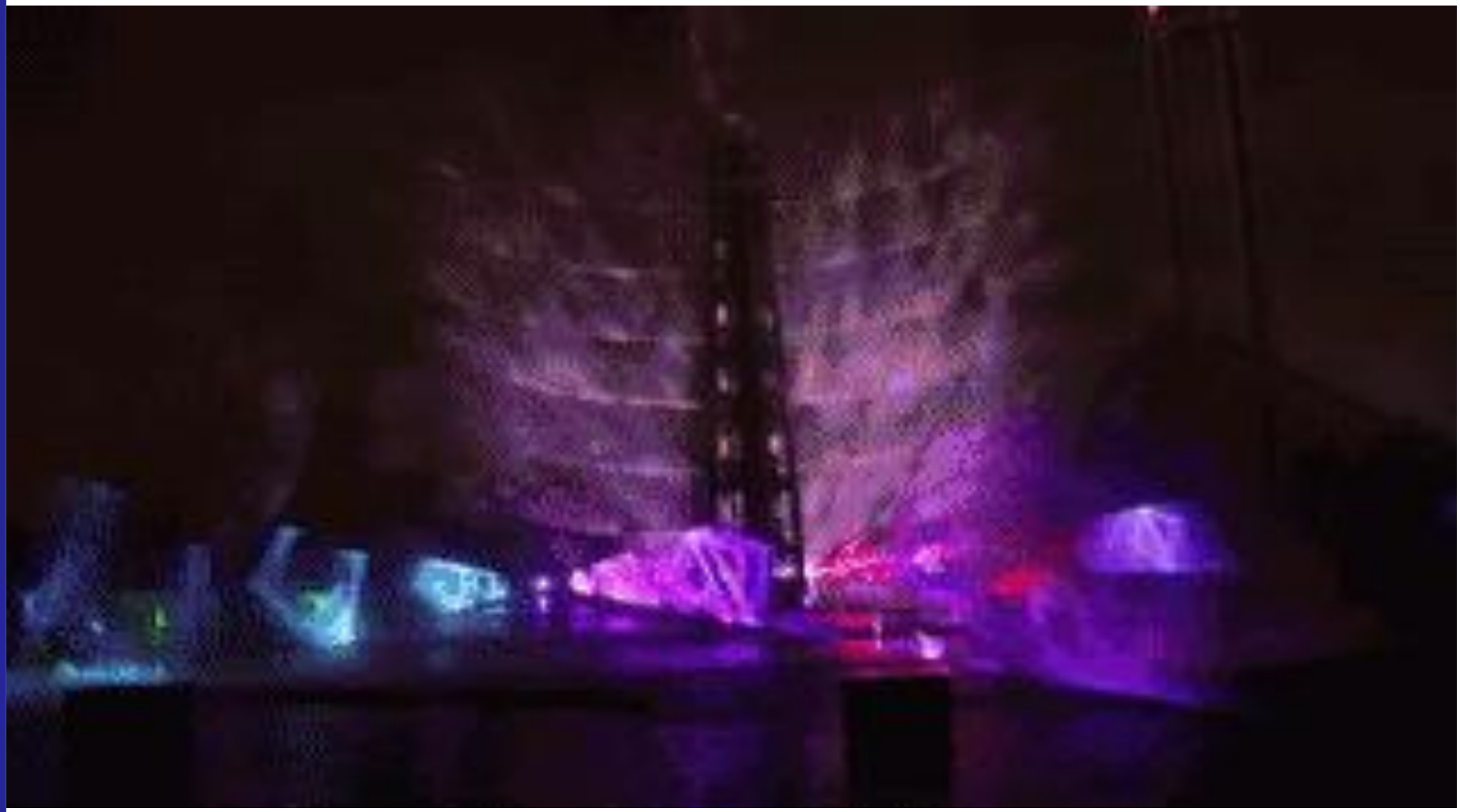
New technology: naked 3D technology





New technology: naked 3D technology






短暂的相遇却念念不忘


MOMENT
FACTORY

2. A parallel beam of light makes an angle of 30° with the surface of a glass having a refractive index of 1.50.

(a) What is the angle between the refracted beam and the surface of glass?

(b) What should be the angle of incidence θ with this plate for the angle of refraction to be $\theta/2$?

4. A narrow beam of light is incident at a 30° angle from a normal onto a glass pane 6mm thick. Describe the position of the exit beam of light. What is the direction? Is it displaced from the incident beam? If so, by how much? (For the glass, $n = 1.60$)

light (Fig. 9-48).

Fig. 9-48 Problem 6.

7. An object located 32cm in front of a lens forms an image

on a screen 8.0cm behind the lens, (a) Find the focal length of the lens, (b) Determine the magnification. (c) Is the lens converging or diverging?

magnification. (c) Is the lens converging or diverging?

8. A converging lens has a focal length of 10cm.For object distances of 30cm, 15cm and 5cm determine, (a) image position, (b) magnification, (c) whether the image is real or virtual, (d) whether the image is erect or inversed.

(b) What is the magnification of the image for these positions of the lens?

10. Two slits are spaced 0.30mm apart and are placed 50cm from a screen. What is the distance between the second and third dark lines of the interference pattern when the slits are illuminated with light of 600nm wavelength?

12. A double-slit experiment produces fringes on a distant screen. How does the linear separation between the bright maxima on the screen change when (a) the wavelength of the light increases? (b) the separation between the slits doubles? (c) the distance between the slits and the screen doubles? (d) the intensity of the light doubles?

13. If mirror M_2 in Michelson interferometer is moved through 0.233mm, 792 fringes are counted with a light meter. What is the wavelength of the light?

14. A thin film with $n = 1.40$ for light of wavelength 589nm is placed in one arm of a Michelson interferometer. If a shift of 7.0 fringes occurs, what is the film thickness?

16. A binary star system in the constellation Orion has an angular separation between the two stars of 1.0×10^{-5}rad. If $\lambda = 500$nm, what is the smallest diameter the telescope can have and just resolve the two stars?

19. A diffraction grating 20mm wide has 6000 rulings.

(a) Calculate the distance d between adjacent rulings.

(b) At what angles will maximum-intensity beams occur if the incident radiation has a wavelength of 589nm?

order diffraction be observed at $\theta = 60°$?

21. Show that, whenever a continuous visible spectrum is passed through a diffraction grating of any spacing size, the third-order fringe of the violet end of thevisible spectrum always overlaps the second-order fringe of the red light at the other end.

22. Monochromatic X-rays are incident on a set of a certain crystal planes whose interplanar spacing is 0.398nm. When the X-ray beam is rotated 60° from the normal, first-order Bragg reflection is observed. What is the wavelength of the X-rays?